Vapor Stone RAIL SYSTEMS
A Wabtec company

10 655 Henri-Bourassa Blvd., West
Saint-Laurent (Québec) Canada
H4S 1A1

Tel: 514.335.4200
Fax: 514.335.4231
www.wabtec.com

Denver Regional Transportation District

Denver EMU Eagle P3 Door System

Project # 7230

Door System

Software Requirement Traceability Matrix

Contract Number	-	CDRL Number	-
		Customer Ref. Number	-

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	Name / Title	Signature / Date
Prepared :	James Kim Software Designer	James Kim 2011-11-07
Verified :	Martin Ouellet, Eng. Software Department Supervisor	2011-11-07
Approved :	Aldo Di Rocco Project Manager	For Aldo Di Rocco 2011-11-07

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Preface

This document is prepared in accordance with IEEE Std. 1558-2004 and Section 4.3.8 of IEEE Std. 830-1998.

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1 INTRODUCTION

1.1 Purpose

This Software Requirements Traceability Matrix (SRTM) document identifies and traces technical requirements allocated to the Door System through their statement of allocation, design delivery and verification of accomplishment, via testing. It is prepared by VSRS for Denver EMU Eagle P3 project.

This SRTM fulfills the specifications of IEEE Std. 1558-2004.

The intended audience of the SRTM is the VSRS software development and maintenance team members, Hyundai Rotem, and Denver Regional Transportation District.

1.2 Scope

This SRTM applies to the SCIs located in the DCU and the PTU Application; which includes:

- DCU Boot Loader Application
- DCU Application
- DCU Neuron Gateway Application
- PTU Application

1.3 Document organization

This STRM is organized in three sections: Introduction, References and Traceability.

Section 1, Introduction, provides an overview of the document activities and audience.

Section 2, References, mentions all referred documents within this SRTM.

Section 3, Traceability, is divided into three sub-sections; each one contains the traceability table with the requirements of each SCI.

Traceability starts with the source document reference (Rotem Specification), which links to the system requirement (SFD); and then it derives to the Software Requirement Specification (SRS), and Software Design Description (SDD) requirements. Validation of the requirements is done in the Software Integration Test Plan (SITP). For each requirement the document sections Sx.x.x (and document identification SDx.x or TCx.x, if applicable) are presented in the tables. The information is reported as follows:

Table 1-1: Software Requirements Traceability Matrix description

SRS Reqmt. ID	Requirements Description (From SRS)	SRS Ref. Section	TS Tag & Section	SFD / System Ref. ID	SDD Specific Ref. ID	SDD Generic Ref. ID	SITP Ref. ID

Specific revision of the mentioned documents is listed in Section 2.

1.4 Definitions and Acronyms

1.4.1 Definitions

Door Control Unit: A processor based electronic unit running software that controls the door, provides monitoring functions, and interacts with the user via a PTU Application and a network.

Event: Term use to globally identify an Info, an Exception or a Failure (see below).

Event Level Definitions By Increasing Importance

Info: An Info is an occurrence of an informative condition monitored by the system, which is neither an exception nor a failure.

Exception: An exception is an occurrence of a condition that is outside of the specified parameters and affecting an outside Line Replaceable Unit (LRU).

Failure: A failure is an occurrence of a condition that is outside of the specified parameters and affecting this LRU. A failure normally requires maintenance or replacement to restore affected unit to its normal operation conditions.

1.4.2 Acronyms

Acronym	Meaning
ASCII	American Standard Code for Information Interchange
CDRL	Contract Data Requirements List
CD-ROM	Compact Disk Read-Only Memory
CPLD	Complex Programmable Logic Device
CRC	Cyclic Redundancy Check
CRC16	16-bit Cyclic Redundancy Check
CRC32	32-bit Cyclic Redundancy Check
CS	Crew Switch
DCU	Door Control Unit
ECD	Engineering Change Document
EDF	Event Description File
ID	Identification
IEEE	Institute of Electrical and Electronics Engineers
ITPS	Integration Test Procedure Specification
LED	Light Emitting Diode
ms	Millisecond
MU	Multiple Units
N/A	Not Applicable
NVRAM	Non-Volatile Random Access Memory
OPDF	Operating Parameters Description File
P/N	Part Number
PIU	Portable Test Equipment
PSS	Panel Sensing Switch
PWM	Pulse Width Modulation
RAM	Random Access Memory
RTDM	Real-Time Data Monitoring
SCI	Software Configuration Item
SCMP	Software Configuration Management Plan
SDD	Software Design Description
SPMP	Software Project Management Plan
SQAP	Software Quality Assurance Plan
SRAM	Static RAM
SRS	Software Requirements Specification
SRTM	Software Requirements Traceability Matrix
SFD	System Functional Description
SVVP	Software Validation and Verification Plan
SW	Switch
TBD	To Be Determined
TL	TrainLine
TS	Technical Specification
USB	Universal Serial Bus
VSRS	Vapor Stone Rail Systems
w/o	without

2 REFERENCES

This section lists the number, title, revision, date, and source of all the documents referenced in this SRTM.

Standards

IEEE Std 1558-2004	IEEE Recommended Practice for Software Documentation for Rail Equipment and Systems, March 23rd 2005, IEEE, www.ieee.org
IEEE Std 830-1998	IEEE Recommended Practice for Software Requirements Specifications, June 25th 1998, IEEE, www.ieee.org

Vapor Stone Rail Systems Documents

201111185	System Functional Description – Door System, Rev. 4
201111186	Software Configuration Management Plan, Door System, Rev. 2
201111207	Software Project Management Plan, Door System, Rev. 1
201111208	Software Quality Assurance Plan, Door System, Rev. 2
201111254	Software Verification and Validation Plan, Door system, Rev. 1
201111256	Software Requirement Specification – Neuron Gateway Application, Rev. 1
201111270	Software Requirement Specification - Generic Door Control Module Application, Rev. 1
201111271	Software Requirements Specification - Generic Boot Loader Application, Rev. 1
201111274	Software Requirements Specification - Generic Portable Test Application, Rev. 1
201111295	Software Design Description - Generic Portable Test Application, Rev. 1
201111450	Master Software Design Description – Door Control Unit Application, Rev. 1
201111453	Software Integration Test Plan - Door Control Unit Application, Rev. 1
201111456	Software Design Description - Generic Boot Loader Application, Rev. 1
201111459	Software Design Description – Neuron Gateway Application, Rev. 1
201111460	Software Integration Test Plan – Neuron Gateway Application, Rev. 1
201111464	Software Integration Test Plan - Portable Test Unit Application, Rev. 1
201111544	Software Design Description – Generic Watchdog Module, Rev. 1
201111554	Software Design Description – Generic Proportional Integral Derivative Module, Rev. 1
201111568	Software Design Description – Generic Door Operator Module, Rev. 1
201111580	Software Integration Test Plan - Generic Boot Loader Application, Rev. 1

201111581	Software Design Description – Generic Real Time Clock Module, Rev. 1
201111582	Software Design Description – Generic Serial Driver Module, Rev. 1
201111583	Software Design Description – Generic Fault Logger Module, Rev. 1
201111584	Software Design Description – Generic Self-Test Module, Rev. 1
201111585	Software Design Description – Generic Real Time Data Monitoring Module, Rev. 1
201111586	Software Design Description – PTA Protocol Module, Rev. 1
201111628	Software Design Description – IO Module, Rev. 1
201111630	Software Design Description – MONIT Module, Rev. 1
201111659	Software Design Description – Neuron SPI Module, Rev. 1

Hyundai Rotem Document

REDE112784	PTS Entrance Door for Denver Eagle P3 Project, Rev. 3, 2011-01-13
REDE113762	System Overview for Side Door System Denver EMU, Rev. 0, 201111-04-18

3 TRACEABILITY

The following sections display the Software Requirements Traceability Matrix for each software requirement.

3.1 DCU Boot Loader Application

Table 3-1 provides a central point of references for the software requirements. It provides a starting point for an upward and downward traceability of all allocated software requirements.

The table refers to the following:

- SRS- Generic Boot Loader Application, Rev. 1, P/N 201111271
  - SRS Requirement ID: Rational RequisitePro Requirement Tag number
  - Requirements Description: SRS requirement text
  - SRS Reference Section: Section number from the SRS document.
- SFD – Door System, Rev. 4, P/N 201111185
  - SFD / System Reference ID: Rational RequisitePro Specification Requirement Tag number
- SDD - Generic Boot Loader Application, Rev. 1, P/N 201111456
  - SDD Generic Reference ID: Rational RequisitePro Design Requirement Tag number
- SITP - Generic Boot Loader Application, Rev. 1, P/N 201111580
  - SITP Reference ID: Rational RequisitePro Test Requirement Tag number

Table 3-1: DCU Boot Loader Application Software Requirements Traceability Matrix

SRS Reqmt ID	Requirements Description (From SRS)	SRS Ref. Section	SFD / System Ref. ID	SDD Generic Ref. ID	SITP Ref. ID
VAP2.1.1	Upon reception of HW_COM.PTA.Message, the Boot Loader software shall verify the integrity of the message content, that the MemoryVolatile.LoginState value is valid (connected) and that the ControllerID of the message corresponds to the ControllerID of the CPU. If these tests are passed, the Boot Loader shall verify the validity of the command contained in the HW_COM.PTA.Message and execute the requested command.	S3.2.1.1		SD1.2.2 SD1.3.1	TC1.1.2 TC1.8.1
VAP2.1.2	If HW_COM.PTA.Message fails one or more of the validity tests, HW_COM.PTA.Message should not be taken into account by the Boot Loader.	S3.2.1.1		SD1.2.2	TC1.2.1 TC1.8.2
VAP2.1.3	Upon reception of an invalid command, the system shall inform the requesting end that the command is not supported.	S3.2.1.1		SD1.3.10	TC1.2.1
VAP2.1.4	Every outgoing HW_COM.BootLoader.Message sent to the PTA by the Boot Loader shall bear the ControllerID of the CPU running the Boot Loader.	S3.2.1.1		SD1.2.1	TC1.8.1
VAP2.2.1	Boot Loader Application shall not accept others commands until the PTA application is logged into Door Module System.	S3.2.1.2		SD1.3.1	TC1.1.1
VAP2.2.2	Upon reception of HW_COM.PTA.LoginRequest from the PTA, the Boot Loader shall update MemoryVolatile.LoginState (MemoryVolatile.LoginState = connected) and answer with HW_COM.BootLoader.LoginResponse	S3.2.1.2	SFD1.1.9.6.1	SD1.3.11 SD1.3.12	TC1.1.8
VAP2.2.3	Upon login, if the PTA is set in Door Module System application mode, the Boot Loader Application shall verify the Door Module System application software integrity and start the Door Module System application.	S3.2.1.2	SFD1.1.9.6.1	SD1.1	TC1.4.1 TC1.4.2

SRS Reqmt ID	Requirements Description (From SRS)	SRS Ref. Section	SFD / System Ref. ID	SDD Generic Ref. ID	SITP Ref. ID
VAP2.3.1	Upon reception of the HW_COM.PTA.LogoutRequest from the PTA, the Boot Loader shall cease communication with the PTA (MemoryVolatile.LoginState = disconnected) and send HW_COM.BootLoader.LogoutResponse.	S3.2.1.3	SFD1.1.9.6.1	SD1.3.13 SD1.3.14	TC1.1.4
VAP2.3.2	Upon reception of HW_COM.PTA.LogoutRequest, the Boot Loader execute the Door Module System application if its integrity is found valid. If it is not valid, the Boot Loader shall continue its execution and wait for a valid login from PTA.	S3.2.1.3	SFD1.1.9.6.1	SD1.3.1 SD1.3.13 SD1.3.14	TC1.1.5
VAP2.4.1	Upon no reception of HW_COM.PTA.Message for more than 1 minute, the Boot loader shall cease communication with the PTA by logging out (MemoryVolatile.LoginState = disconnected).	S3.2.1.4		SD1.3.1	TC1.1.6 TC1.1.7
VAP2.5.1	Every 30 seconds, the PTA shall send HW_COM.PTA.HeartbeatRequest to the Boot Loader and receive HW_COM.BootLoader.HeartbeatResponse in answer, to ensure that a working connection is still established between the Boot Loader and the PTA.	S3.2.1.5		SD1.3.17 SD1.3.18	TC1.10.1
VAP2.6.1	Upon reception of HW_COM.PTA.SoftwareVersionsRequest from the PTA, the Boot Loader software shall send HW_COM.BootLoader.SoftwareVersionsResponse containing MemoryApplication.BootLoaderApplication.AppInfo and MemoryApplication.DMSApplication.AppInfo in answer.	S3.2.1.6	SFD1.1.9.6.11	SD1.3.4 SD1.3.5	TC1.5.1
VAP2.7.1	Upon reception of HW_COM.PTA.CodeStatusRequest, the Boot Loader shall send HW_COM.BootLoader.CodeStatusResponse containing The CRC and the size of the DMS and Boot Loader applications to the PTA Application.	S3.2.1.7	SFD1.1.9.6.11	SD1.3.15 SD1.3.16	TC1.6.1
VAP2.8.1	Upon reception of HW_COM.PTA.DMSApplicationUpgradeRequest, the Boot Loader shall receive the DMS application code and test its integrity. If successful, the Boot Loader shall then program it into MemoryApplication.DMSApplication and send HW_COM.BootLoader.DMSApplicationUpgradeResponse.	S3.2.1.8		SD1.3.2 SD1.3.3 SD1.3.6 SD1.3.7 SD1.3.8 SD1.3.9 SD1.8	TC1.4.1 TC1.4.2
VAP2.9.1	The system shall allow upgrade of the Boot Loader application through the HW_COM interface into MemoryApplication.BootLoaderApplication. The implementation of this function is specific to the CPU being used and shall be define in the design phase.	S3.2.1.9		SD1.3.2 SD1.3.3 SD1.3.6 SD1.3.7 SD1.3.8 SD1.3.9 SD1.8	TC1.7.1 TC1.7.2
VAP2.10.1	Upon the reception of MemDumpRequest from the PTA, the Boot Loader shall transfer the memory content.	S3.2.1.10		SD1 SD1.3.2 SD1.3.3 SD1.3.6 SD1.3.7 SD1.3.8 SD1.3.9	TC1.13.1
VAP2.11.1	The system shall verify the integrity of MemoryApplication.DMSApplication and MemoryApplication.BootLoaderApplication at boot-up and set the corresponding MemoryApplication.BootLoaderApplication.IntegrityError or MemoryApplication.DMSApplication.IntegrityError.	S3.2.2.1	SFD1.1.9.2.2	SD1.7	TC1.4.1 TC1.4.2
VAP2.11.2	The Boot Loader shall stay in boot loader mode whenever the integrity of DMS application is not valid and MemoryApplication.DMSApplication.IntegrityError shall be activated.	S3.2.2.1	SFD1.1.9.2.2	SD1.7	TC1.3.1 TC1.3.2 TC1.3.3
VAP2.12.1	The Boot Loader shall verify the integrity of MemoryVolatile.Content before it can use MemoryVolatile	S3.2.2.2	SFD1.1.9.2.2 SFD1.1.9.2.3	SD1.6.1 SD1.6.2	TC1.3.1
VAP2.12.2	If an error occurs during MemoryVolatile.Content test, MemoryVolatile.IntegrityError shall be activated.	S3.2.2.2	SFD1.1.9.2.2 SFD1.1.9.2.3	SD1.6.1 SD1.6.2	TC1.3.2
VAP2.13.1	The Boot Loader shall verify the integrity of MemoryNonVolatile.Content before it can use MemoryNonVolatile	S3.2.2.3	SFD1.1.9.2.2 SFD1.1.9.2.3	SD1.6.2	TC1.3.1
VAP2.13.2	If an error occurs during MemoryNonVolatile.Content test, MemoryNonVolatile.IntegrityError shall be activated.	S3.2.2.3	SFD1.1.9.2.2 SFD1.1.9.2.3	SD1.6.2	TC1.3.3

SRS Reqmt ID	Requirements Description (From SRS)	SRS Ref. Section	SFD / System Ref. ID	SDD Generic Ref. ID	SITP Ref. ID
VAP2.13.3	MemoryNonVolatile.Content shall be preserved from corruption during integrity test	S3.2.2.3		SD1.6.2	TC1.3.3
VAP2.14.1	The system shall blink the LEDWarning while the system is in the Boot Loader. The retained blinking pattern should be unique.	S3.2.3.1		SD1.5	TC1.3.1
VAP2.14.2	If MemoryNonVolatile.IntegrityError is activated, the system shall blink the LEDFault using LEDFault.MemoryNonVolatileIntegrityError unique pattern	S3.2.3.1		SD1.5	TC1.3.3
VAP2.14.3	If MemoryVolatile.IntegrityError is activated, the system shall blink the LEDFault using LEDFault.MemoryVolatileIntegrityError unique pattern.	S3.2.3.1		SD1.5	TC1.3.2
VAP2.14.4	For each error to be displayed by the LEDFault, the system should use an unique pattern identifying it unambiguously	S3.2.3.1		SD1.5	TC1.3.1 TC1.3.2 TC1.3.3
VAP2.14.5	If MemoryApplication.DMSApplication.IntegrityError is activated, the system shall blink the LEDFault using LEDFault.DMSApplicationIntegrityError unique pattern	S3.2.3.1		SD1.5	TC1.4.1
VAP2.14.6	The system shall blink the LEDFault in a unique pattern if MemoryApplication.BootLoaderApplication.IntegrityError is activated.	S3.2.3.1		SD1.5	TC1.3.1
VAP2.15.1.1	Boot Loader shall use less than 10% of each installed memory.	S3.3.1		SD1.10.1	TC1.9.1
VAP2.15.1.2	Boot Loader shall use less than 50% of available processing time.	S3.3.1		SD1.9.1	TC1.9.2
VAP2.16.1.1	The Boot Loader shall be fully functional in less than 60 seconds after a power-up or a reset	S3.6.2		SD1.1	TC1.9.3
VAP2.16.1.2	To ensure the system availability, the boot Loader's design should include provisions for error detection and recovery, both software and hardware.	S3.6.2		SD1.6 SD1.6.1 SD1.6.2 SD1.7	TC1.9.4
VAP2.16.2.1	The system shall not allow access to any door control in the Boot Loader.	S3.6.3		SD1.1	TC1.11.1
VAP2.16.3.1	The Boot Loader shall be upgraded by the Portable Test Application through USB chip that drives 2 I/O, or if a jumper is installed on a special connector on Door Module System electronic card while it is rebooting.	S3.6.4		SD1.3.1	TC1.7 TC1.7.1

3.2 DCU Application

Table 3-2 provides a central point of references for the software requirements. It provides a starting point for an upward and downward traceability of all allocated software requirements.

The table refers to the following:

- SRS - Generic Door Control Module Application, Rev. 1, P/N 201111270
  - SRS Requirement ID: Rational RequisitePro Requirement Tag number
  - Requirements Description: SRS requirement text
  - SRS Reference Section: Section number from the SRS document.
- Rotem Technical Specification and Section
  - RTS: Rational RequisitePro Specification Requirement Tag number for Rotem Technical Specification
  - SO Section: Section of the System Overview for Side Door System Denver EMU P/N REDE113762
  - PTS Section: Section of the Procurement Technical Specification of Entrance Door System P/N REDE112784
- SFD – Door System, Rev. 4, P/N 201111185
  - SFD / System Reference ID: Rational RequisitePro Specification Requirement Tag number
- SDD Specific: Rational RequisitePro Design Requirement Tag number
  - Rational RequisitePro Tag starts with SD98: SDD - IO Module, Rev. 1, P/N 201111628
  - Rational RequisitePro Tag starts with SD100: SDD - MONIT Module, Rev. 1, P/N 201111630
- SDD Generic: Rational RequisitePro Design Requirement Tag number
  - Rational RequisitePro Tag starts with SD2: SDD - DOP Module, Rev. 1, P/N 201111568
  - Rational RequisitePro Tag starts with SD3: SDD - Fault Module, Rev. 1, P/N 201111583
  - Rational RequisitePro Tag starts with SD5: SDD - PTA Module, Rev. 1, P/N 201111586
  - Rational RequisitePro Tag starts with SD6: SDD - Real-Time Clock Module, Rev. 1, P/N 201111581
  - Rational RequisitePro Tag starts with SD7: SDD - RTDM Module, Rev. 1, P/N 201111585
  - Rational RequisitePro Tag starts with SD8: SDD - Self-Test Module, Rev. 1, P/N 201111584
  - Rational RequisitePro Tag starts with SD10: SDD - Neuron SPI Module, Rev. 1, P/N 201111659
- SITP - Door Control Unit Application, Rev. 1, P/N 201111453
  - SITP Reference ID: Rational RequisitePro Test Requirement Tag number

Table 3-2: DCU Application Software Requirements Traceability Matrix

SRS Reqmt. ID	Requirements Description (From SRS)	SRS Ref. Section	TS Tag & Section	SFD / System Ref. ID	SDD Specific Ref. ID	SDD Generic Ref. ID	SITP Ref. ID
VAP1.1.1	On system startup and continuously, at a slow rate (one test per second) and with a minimum of one test per day, DCM application shall execute the self-test routine, testing the MemoryVolatile and MemoryNonVolatile, the DCM and Boot loader application integrities and the CGM communication status.	S3.2.1.1		SFD1.1.9.2.3		SD8.1 SD8.2	TC1.1.1.1
VAP1.1.1.1	MemoryVolatile and MemoryNonVolatile	S3.2.1.1		SFD1.1.9.2.1		SD8.2	TC1.1.1.2
VAP1.1.1.1.1	The self-test shall verify that MemoryVolatile and MemoryNonVolatile can read and write data without generating any error.	S3.2.1.1		SFD1.1.9.2.1		SD8.1 SD8.4	TC1.1.1.2

SRS Reqmt. ID	Requirements Description (From SRS)	SRS Ref. Section	TS Tag & Section	SFD / System Ref. ID	SDD Specific Ref. ID	SDD Generic Ref. ID	SITP Ref. ID
VAP1.1.1.1.2	The self-test shall verify the integrity of MemoryVolatile.Content and MemoryNonVolatile.Content.	S3.2.1.1		SFD1.1.9.2.1		SD8.1 SD8.2 SD8.5	TC1.1.1.2
VAP1.1.1.2.1	The self-test shall check the integrity of the MemoryApplication.DCMApplication.	S3.2.1.1		SFD1.1.9.2.2		SD8.1 SD8.2 SD8.3	TC1.1.1.3
VAP1.1.1.3.1	The self-test shall check the integrity of the MemoryApplication.BootLoaderApplication.	S3.2.1.1				SD8.1 SD8.2 SD8.3	TC1.1.1.4
VAP1.1.1.4.1	The self-test shall verify CGM.CommunicationStatus with the CGM application.	S3.2.1.1				SD8.1 SD8.2	TC1.1.1.5
VAP1.1.2	The result of self-tests shall be stored as MemoryVolatile.SelfTestResults to be retrieved by PTA or sent to the CGM to update the Car Network information.	S3.2.1.1		SFD1.2		SD8.1 SD8.2	TC1.1.1.5
VAP1.1.3	All failures detected during a self-test shall be logged in the MemoryNonVolatile.EventLogFile.	S3.2.1.1		SFD1.1.9.6.6		SD8.1 SD8.2	TC1.1.1.6
VAP1.1.4	All memories failures shall reset the DCM Application.	S3.2.1.1				SD8.1 SD8.2	TC1.1.1.7
VAP1.2.1	The DCM application shall monitor all inputs signal values listed in section 2.1.1 and system components in order to identify any unusual Info, Exception or Failure as defined in the project's specific Fault Management Plan (FMP).	S3.2.1.2			SD100.11		TC1.1.1.8 TC1.1.3.1 TC1.1.4.2
VAP1.2.2	If an Info or Failure is found, the MemoryNonVolatile.EventStatus shall be updated accordingly and it shall be logged in MemoryNonVolatile.EventLogFile.	S3.2.1.2				SD3.4	TC1.1.4.2
VAP1.2.3	If an Info or Failure disappears, the MemoryNonVolatile.EventStatus shall be updated accordingly and it shall be logged in MemoryNonVolatile.EventLogFile.	S3.2.1.2				SD3.4	TC1.1.4.3
VAP1.2.4	The DCU Application shall set the flag MemoryVolatile.MODE_DEGRADED. DisableOpen when any one of "set" Events has attribute of "Disable Open". The DCU Application shall reset the flag MemoryVolatile.MODE_DEGRADED.DisableOpen when there are no "set" Events with "Disable Open" attribute.	S3.2.1.2				SD2.6	TC1.1.4.5
VAP1.2.5	The DCU Application shall set the flag MemoryVolatile.MODE_DEGRADED. DisableClose when any one of "set" Events has attribute of "Disable Close". The DCU Application shall reset the flag MemoryVolatile.MODE_DEGRADED.DisableClose when there are no "set" Events with "Disable Close" attribute.	S3.2.1.2				SD2.7	TC1.1.4.6
VAP1.2.6	The DCU Application shall set the flag MemoryVolatile.MODE_DEGRADED. ReducedSpeed when any one of "set" Events has attribute of "Reduced Speed". The DCU Application shall reset the flag MemoryVolatile.MODE_DEGRADED.ReducedSpeed when there are no "set" Events with "Reduced Speed" attribute.	S3.2.1.2				SD2.6 SD2.7	TC1.1.4.7
VAP1.3.1	Upon activation of a valid open command, DCM application shall initiate the door open sequence corresponding to the Input.MotorTypeID.ISO.	S3.2.2.1	RTS2.1.1 SO Section 3.2.1: Driver's Mode	SFD1.1.1.1		SD2.1 SD2.6	TC1.1.6.2
VAP1.3.2	To open the door using the Outside Crew Switch, the DCM Application shall detect a valid Outside Crew Switch Open command.	S3.2.2.1			SD100.14		TC1.1.6.5
VAP1.3.3	To open the door using the Inside Crew Switch, the DCM Application shall detect shall detect a valid Inside Crew Switch Open command. **(Not applicable to Denver Project)**	S3.2.2.1			N/A	N/A	N/A
VAP1.3.4	The door shall open in the defined time (SACS.Door.OpeningTime) by default.	S3.2.2.1				SD2.6	TC1.1.6.2
VAP1.3.5	If Input.DO.MotorTemp is measured and over the temperature limit, MemoryVolatile. MODE_DEGRADED.ReduceSpeed shall be set and the low current limit Error shall be activated. **(Not applicable to Denver Project)**	S3.2.2.1			N/A	N/A	N/A
VAP1.3.6	If a solenoid lock is present, the DCM application shall first drive the motor in close direction to remove any pressure and unlock the door by driving the solenoid lock.	S3.2.2.1				SD2.6 SD2.9	TC1.1.6.2

SRS Reqmt. ID	Requirements Description (From SRS)	SRS Ref. Section	TS Tag & Section	SFD / System Ref. ID	SDD Specific Ref. ID	SDD Generic Ref. ID	SITP Ref. ID
VAP1.3.7	The door shall be maintained in the fully open position after opening.	S3.2.2.1				SD2.6	TC1.1.6.2
VAP1.3.8	The MemoryNonVolatile.DoorMovementinformationFiles for opening shall include door speed, PWM motor command, and position error.	S3.2.2.1				SD2.6	TC1.1.11.7 TC1.1.25.1
VAP1.3.9	The DCM Application shall open doors within tolerance of ±100ms from the configured opening time (SACS.Door.OpeningTime).	S3.2.2.1				SD2.6	TC1.1.6.3
VAP1.3.10	The DCM Application shall open doors with maximum opening time (SACS.Door.OpeningTime.Range) when the MemoryVolatile.MODE_DEGRADED. ReduceSpeed is set.	S3.2.2.1				SD2.6	TC1.1.6.3
VAP1.3.11	The DCM Application shall not open doors when the MemoryVolatile.MODE_DEGRADED.DisableOpen flag is set	S3.2.2.1				SD2.6	TC1.1.4.5
VAP1.4.1	When the door is not already closed and locked, and the DCM receive a valid close command; the DCM application shall activate the door closing warning indicator for the SACS.Door.ClosingDelay delay and initiate the door close sequence corresponding to the Input.MotorTypeID.ISO.	S3.2.2.2		SFD1.1.1.2 SFD1.1.1.3 SFD1.1.1.3.2 SFD1.1.1.4 SFD1.1.1.5		SD2.3 SD2.7	TC1.1.7.5
VAP1.4.2	To close the door using the Outside Crew Switch, the DCM Application shall detect a valid Outside Crew Switch Close command.	S3.2.2.2			SD100.14		TC1.1.7.6
VAP1.4.3	To close the door using the Inside Crew Switch, the DCM Application shall detect a valid Inside Crew Switch Close command.	S3.2.2.2			SD100.14		TC1.1.7.6
VAP1.4.4	The DCM Application shall flash the Output.PassAlertLight.ISO (If SACS.Door.VisualAlarmActivation is activated) and Output.PassAudibleSignal.ISO (If SACS.Door.AudibleAlarmActivation is activated) only when a valid Close command is detected.	S3.2.2.2	RTS2.3.1 SO Section 3.4.1: Audible Alarm	SFD1.1.7.1	SD98.9	SD2.7	TC1.1.7.1
VAP1.4.5	The door shall close according to configured time (SACS.Door.ClosingTime).	S3.2.2.2				SD2.7	TC1.1.7.1
VAP1.4.6	The MemoryNonVolatile.DoorMovementinformationFiles for closing shall include door speed, PWM motor command, and position error.	S3.2.2.2				SD2.7	TC1.1.11.6 TC1.1.11.7 TC1.1.25.2
VAP1.4.7	Upon closure of the door, the DCM shall increment the door cycle counter within the MemoryNonVolatile.SystemCountersFile.	S3.2.2.2				SD2.7	TC1.1.32.1
VAP1.4.8	The DCM application shall close doors within tolerance of ±100ms from the configured closing time (SACS.Door.ClosingTime).	S3.2.2.2				SD2.7	TC1.1.7.1
VAP1.4.9	If the door has never been locked since last power-up, MemoryVolatile.PositionEncoderCount shall be considered unknown and the door shall close slowly, in seek-home mode.	S3.2.2.2		SFD1.1.1.5		SD2.5 SD2.7 SD2.8 SD2.12 SD2.13 SD2.15	TC1.1.7.2
VAP1.4.10	The DCM Application shall close doors using the maximum closing time (SACS.Door.ClosingTime.Range) when the MODE.DEGRADED flag “Reduced Speed” is set.	S3.2.2.2				SD2.7	TC1.1.4.7
VAP1.4.11	The DCM Application shall not close doors when the MODE.DEGRADED flag “Disable Close” is set	S3.2.2.2				SD2.7	TC1.1.4.6
VAP1.4.12	If Input.DO.MotorTemp is over the limit, the MODE.DEGRADED flag “Reduced Speed” shall be set to minimize the load on the motor that is already overheating.	S3.2.2.2				SD2.7	
VAP1.5.1	DCM application shall detect a flat bar, 3/8-inch (9.53 mm) thick and 3 inches (76.2 mm) wide, held rigidly between and perpendicular to the door panel, as a hand might be held to stop the doors.	S3.2.2.3	RTS2.2.3 SO Section 3.3.3: Obstrucion Detection	SFD1.1.2.1.1		SD2.2 SD2.7	TC1.1.8.1

SRS Reqmt. ID	Requirements Description (From SRS)	SRS Ref. Section	TS Tag & Section	SFD / System Ref. ID	SDD Specific Ref. ID	SDD Generic Ref. ID	SITP Ref. ID
VAP1.5.2	This sensitivity shall be required throughout the entire length of the panel and for both values of the Input.MotorTypeID.ISO.	S3.2.2.3	RTS2.2.3 SO Section 3.3.3: Obstrucion Detection	SFD1.1.2.1.1		SD2.2 SD2.7	TC1.1.8.1
VAP1.5.3	It shall also detect an object, 7/16-inch (11.11 mm) in diameter, held rigidly between and perpendicular to the door panels at all locations along the length of the door nosing seal for all motor types supported, except the uppermost and lowermost 4 inch (101.6 mm) of the seal.	S3.2.2.3	RTS2.2.3 SO Section 3.3.3: Obstrucion Detection	SFD1.1.2.1.2		SD2.2 SD2.7	TC1.1.8.2
VAP1.5.4	When DCM application is closing the door, if the DCM detects an obstruction i.e. a discrepancy between current MemoryVolatile.PositionEncoderCount (calculated by identifying the sequence of Input.DO.MotorEncPhaseA and Input.DO.MotorEncPhaseB values) and the calculated profile position, DCM application shall control the motor to stop, then if the door has not reached the defined obstruction retries limit (SACS.Door.Obstruction.Count), the door shall open by SACS.Door.Obstruction.Distance and retry to close.	S3.2.2.3		SFD1.1.2.1.3 SFD1.1.2.2.1		SD2.2 SD2.4 SD2.7 SD2.10	TC1.1.8.3
VAP1.5.5	When the defined maximum number of retries (SACS.Door.Obstruction.Count) is exceeded, the DCM application shall follow the end behavior selected by SACS.Door.Obstruction.EndBehavior value into the list of 3 pre-defined behavior : 0 Stand Still: the door does not move until the next received command. 1 Fully-Reopen: The door fully reopen at the minimum speed. 2 Force Close: The door keep driving in the close direction.	S3.2.2.3		SFD1.1.2.3.4		SD2.2 SD2.7 SD2.15	TC1.1.8.4
VAP1.5.6	When SACS.Door.Obstruction.EndBehavior is in Force Close mode, the current in the motor shall be limited to avoid temperature build up in the motor	S3.2.2.3					TC1.1.18 TC1.1.18.1
VAP1.6.1	When the Input.DO.EmerHandleSwitch.ISO is activated the DCU Application shall cut the motor power (CPU.PWM.MOTOR), causing the door to be free to move manually.	S3.2.3		SFD1.1.4.3	SD100.15	SD2.14	TC1.1.5.11 TC1.1.18.1
VAP1.7.1	When Input.DPCutoutSwitchR.ISO or Input.DPCutoutSwitchL.ISO is activated the DCM Application shall keep the corresponding door locked (Output.DO.SolenoidDrv = Off) and ignore further command until the Cutout switch is returned to its OFF position.	S3.2.4			SD100.16		TC1.1.5.9 TC1.1.18.1
VAP1.8.1	Upon detection of an event, the DCM application shall store the specific event information along with a timestamp in the MemoryNonVolatile.EventLogFile to prevent information loss resulting from a power failure.	S3.2.5		SFD1.1.9.3.1 SFD1.1.9.3.5 SFD1.1.9.3.7		SD3.4 SD3.6	TC1.1.11.1 TC1.2.1.3.1
VAP1.8.2	The MemoryNonVolatile.EventLogFile shall be able to store 1500 entries.	S3.2.5				SD3.4	TC1.1.11.3 TC1.2.1.3.1
VAP1.8.3	A maximum of 10 entries per event number shall be stored for a single day.	S3.2.5		SFD1.1.9.3.3		SD3.4	TC1.1.11.2 TC1.2.1.3.1
VAP1.8.4	In case of an overflow of the MemoryNonVolatile.EventLogFile, the new data entries shall replace the entries that are stored at the beginning of the MemoryNonVolatile.EventLogFile.	S3.2.5				SD3.4	TC1.1.11.4 TC1.2.1.3.1
VAP1.9.1	Each snapshot sample is a single set of values for a specified group of signals, a snapshot sample shall be taken every SACS.Door.Snapshot.Period.	S3.2.6				SD7.3	TC1.1.11.5
VAP1.9.2	The SACS.Door.Snapshot.Capacity most recent snapshot samples are stored in a circular buffer in the MemoryVolatile.SnapshotBuffer to keep the last (SACS.Door.Snapshot.Capacity * SACS.Door.Snapshot.Period)/1000 seconds status.	S3.2.6		SFD1.1.9.4.1		SD7.3	TC1.1.11.5
VAP1.9.3	When a event with the snapshot attribute occurs, the set of snapshot samples located in the MemoryVolatile.TemporaryData, the next SACS.Door.Snapshot. SampleNbAfterevent snapshots, the current sample and the information identifying the event shall be stored as a MemoryNonVolatile.SnapshotFile.	S3.2.6		SFD1.1.9.4.2 SFD1.1.9.4.3		SD7.3	TC1.1.11.5

SRS Reqmt. ID	Requirements Description (From SRS)	SRS Ref. Section	TS Tag & Section	SFD / System Ref. ID	SDD Specific Ref. ID	SDD Generic Ref. ID	SITP Ref. ID
VAP1.10.1	Upon reception of HW_COM.PTA.Message, the DCM application shall verify the integrity of the message content, that the MemoryVolatile.LoginState is connected and that the Input.DeviceID of the message corresponds to the Controller ID of the CPU. If HW_COM.PTA.Message fails one or more of the validity tests, HW_COM.PTA.Message should not be taken into account by the DCM application.	S3.2.7.1				SD5.1 SD5.6 SD5.25	TC1.1.30.1
VAP1.10.2	Before sending HW_COM.DMS.Message, the DCM shall include the integrity check result and the Input.DeviceID value of its CPU into the message.	S3.2.7.1				SD5.1 SD5.25	TC1.1.30.1
VAP1.11.1	Upon reception of the HW_COM.PTA.LoginRequest from the PTA, the DCM application shall send HW_COM.DMS.LoginResponse, set MemoryVolatile.LoginState to Connected and start accepting other PTA commands.	S3.2.7.2		SFD1.1.9.6.1		SD5.15 SD5.16	TC1.1.12.1
VAP1.12.1	Upon reception of the HW_COM.PTA.LogoutRequest command, the DCM application shall send HW_COM.DMS.LogoutResponse, set the MemoryVolatile. LoginState to Disconnected and wait for the PTA to login to accept any other PTA command.	S3.2.7.3		SFD1.1.9.6.1		SD5.17 SD5.18	TC1.1.13.1
VAP1.12.2	Upon reception of the HW_COM.PTA.RebootRequest command or a watchdog reset, the DCM application shall discontinue all operations and perform a reboot.	S3.2.7.3				SD5.17	TC1.1.13.2
VAP1.13.1	Upon no reception of HW_COM.PTA.Message for more than 1 minute while MemoryVolatile.LoginState = Connected, the DCM shall cease communication with the PTA by logging out (MemoryVolatile.LoginState = Disconnected).	S3.2.7.4		SFD1.1.9.6.1		SD5.25	TC1.1.13.1
VAP1.14.1	Every 30 seconds, the PTA send HW_COM.PTA.HeartbeatRequest to the DCM and the DCM shall send HW_COM.DMS.HeartbeatResponse to the PTA in answer, to ensure that a working connection is still established between the DCM and the PTA.	S3.2.7.5				SD5.25	TC1.1.30.1
VAP1.15.1	Upon reception of the HW_COM.PTA.LocationIdentificationRequest from the PTA, the DCM application shall send HW_COM.DMS.LocationIdentificationResponse message to the PTA containing MemoryVolatile.LocationIdentification.	S3.2.7.6		SFD1.1.9.6.10		SD5.13 SD5.14	TC1.1.14.1
VAP1.16.1	Upon reception of the HW_COM.PTA.CarNumberRequest from the PTA, the DCM application shall send HW_COM.DMS.CarNumberResponse message to the PTA containing MemoryVolatile.CarNumber.	S3.2.7.7		SFD1.1.9.6.10		SD5.13 SD5.14	TC1.1.14.1
VAP1.17.1	Upon reception of the HW_COM.PTA.LocalTimeRequest from the PTA, the DCM application shall send HW_COM.DMS.LocalTimeResponse containing the current MemoryVolatile.LocalTime value to the PTA.	S3.2.7.8		SFD1.1.9.6.8		SD5.28 SD5.29 SD6.1	TC1.1.15.1
VAP1.18.1	Upon reception of the HW_COM.PTA.SetTimeRequest from the PTA, the DCM application shall send HW_COM.DMS.SetTimeResponse and change MemoryVolatile.LocalTime with the value received from the PTA.	S3.2.7.9		SFD1.1.9.6.8		SD5.28 SD5.29 SD6.2	TC1.1.15.1
VAP1.19.1	Upon reception of the HW_COM.PTA.SelfTestRequest from the PTA, the DCM application shall send HW_COM.DMS.SelfTestResponse to the PTA with last test results.	S3.2.7.10				SD5.22 SD5.23	TC1.1.16.1
VAP1.20.1	Upon reception of the HW_COM.PTA.CodeStatusRequest from the PTA, the DCM application shall send HW_COM.DMS.CodeStatusResponse containing MemoryNonVolatile.DCMApplicationIntegrityCheck, MemoryNonVolatile.BootLoader ApplicationIntegrityCheck, MemoryNonVolatile.DCMApplicationCodeSize, MemoryNon Volatile.BootLoaderApplicationCodeSize to the PTA.	S3.2.7.11				SD5.2 SD5.3	TC1.1.17.1
VAP1.21.1	Upon reception of HW_COM.PTA.MonitoringRequest from the PTA, the DCM application shall send HW_COM.DMS.MonitoringResponse, transmitting all inputs signal values and all outputs signal values listed in section 2.1.1 and defined in the OPDF.	S3.2.7.12		SFD1.1.9.6.5		SD5.32 SD5.33	TC1.1.18.1
VAP1.22.1	Upon reception of the HW_COM.PTA.EventStatusRequest from the PTA, the DCM application shall send HW_COM.DMS.EventStatusResponse containing the MemoryNonVolatile.EventStatus as defined in the EDF.	S3.2.7.13		SFD1.1.9.6.2		SD5.7 SD5.8 SD5.11 SD5.12 SD5.19 SD5.20 SD5.26 SD5.27	TC1.1.19.1

SRS Reqmt. ID	Requirements Description (From SRS)	SRS Ref. Section	TS Tag & Section	SFD / System Ref. ID	SDD Specific Ref. ID	SDD Generic Ref. ID	SITP Ref. ID
VAP1.23.1	Upon reception of HW_COM.PTA.EraseEventLogFileRequest from the PTA, the DCM application shall erase all entries in MemoryNonVolatile.EventLogFile and send HW_COM.DMS.EraseEventLogFileResponse.	S3.2.7.14		SFD1.1.9.6.2		SD3.1 SD3.5 SD5.9 SD5.10	TC1.1.20.1
VAP1.24.1	Upon reception of HW_COM.PTA.EraseSnapshotFileRequest with a snapshot file index argument from the PTA, the DCM application shall erase the selected snapshot file in MemoryNonVolatile.SnapshotFile and send HW_COM.DMS.Erase SnapshotFileResponse.	S3.2.7.14		SFD1.1.9.6.3		SD5.9 SD5.10 SD7.4	TC1.1.21.1
VAP1.25.1	Upon reception of HW_COM.PTA.EventLogFileRequest from the PTA, the DCM application shall send HW_COM.DMS.EventLogFileResponse containing all active entries of MemoryNonVolatile.EventLogFile to the PTA.	S3.2.7.16				SD3.2 SD3.3 SD5.26 SD5.27	TC1.1.22.1
VAP1.26.1	Upon reception of HW_COM.PTA.SnapshotFileRequest from the PTA, the DCM application shall send HW_COM.DMS.SnapshotFileResponse to the PTA, containing MemoryNonVolatile.SnapshotFile.	S3.2.7.17		SFD1.1.9.6.3		SD5.7 SD5.8 SD5.11 SD5.12 SD5.19 SD5.20 SD7.1 SD7.2	TC1.1.23.1
VAP1.27.1	Upon reception of the HW_COM.PTA.EventDescriptionFileRequest from the PTA, the DCM application shall send HW_COM.DMS.EventDescriptionFileResponse containing the MemoryNonVolatile.EventDescriptionFile.	S3.2.7.18				SD5.7 SD5.8 SD5.11 SD5.12 SD5.19 SD5.20	TC1.1.24.1
VAP1.27.2	Upon reception of the HW_COM.PTA.OperatingParametersDescription FileRequest, the DCM application shall send HW_COM.DMS.OperatingParameters DescriptionFileResponse containing the MemoryNonVolatile.OperatingParameters DescriptionFile.	S3.2.7.18				SD5.7 SD5.8 SD5.11 SD5.12 SD5.19 SD5.20	TC1.1.24.2
VAP1.28.1	Upon reception HW_COM.PTA.DoorMovementInformationFilesRequest, the DCM application shall send HW_COM.DMS.DoorMovementInformationFilesResponse containing MemoryNonVolatile.DoorMovementInformationFiles.	S3.2.7.19		SFD1.1.9.6.7		SD5.7 SD5.8 SD5.11 SD5.12 SD5.19 SD5.20	TC1.1.25.1 TC1.1.25.2
VAP1.29.1	Upon reception of a HW_COM.PTA.SoftwareVersionsRequest, the DCM application shall send HW_COM.DMS.SoftwareVersionsResponse containing the MemoryNonVolatile.DCMApplicationVersion, and MemoryNonVolatile.BootLoader ApplicationVersion.	S3.2.7.20		SFD1.1.9.6.11		SD5.21 SD5.30 SD5.31	TC1.1.27.1 TC1.1.27.2
VAP1.30.1	Upon reception of HW_COM.PTA.MemoryDumpRequest from the PTA, the DCM shall send HW_COM.DMS.MemoryDumpResponse in return, containing the MemoryVolatile.Content portion defined by the starting address and the end address of the memory zone to download transmitted by the PTA.	S3.2.7.21				SD5.7 SD5.8 SD5.11 SD5.12 SD5.19 SD5.20	TC1.1.26.1
VAP1.31.1	1 Upon reception of the HW_COM.PTA.ReadDMSConfigurationRequest, the DCM application shall send HW_COM.DMS.DMSConfigurationResponse containing all configurable parameters listed in Appendix A.	S3.2.7.22				SD5.4 SD5.5	TC1.1.28.1 TC1.1.28.2
VAP1.31.2	Available parameters and their read/write associated permissions shall be as defined in the Appendix A.	S3.2.7.22				SD2.16	TC1.1.28.1 TC1.1.28.2

SRS Reqmt. ID	Requirements Description (From SRS)	SRS Ref. Section	TS Tag & Section	SFD / System Ref. ID	SDD Specific Ref. ID	SDD Generic Ref. ID	SITP Ref. ID
VAP1.32.1	Upon reception of the HW_COM.PTA.WriteDMSConfigurationRequest, the DCM application shall update the user configurable parameters in the Configuration File (See Appendix A for description of each parameter) and send HW_COM.DMS.WriteDMSConfigurationResponse.	S3.2.7.23				SD2.16 SD5.4 SD5.5	TC1.1.28.1 TC1.1.28.2
VAP1.32.2	Opening time shall be adjustable within the defined range (SACS.Door.OpeningTime.Range) for all motor types supported.	S3.2.7.23				SD2.6	TC1.1.28.1 TC1.1.28.2
VAP1.32.3	Closing time shall be adjustable into the defined range (SACS.Door.ClosingTime.Range) for both motor types supported.	S3.2.7.23				SD2.7	TC1.1.28.1 TC1.1.28.2
VAP1.32.4	Obstruction retries shall be adjustable within the defined range (SACS.Door.Obstruction.Range).	S3.2.7.23				SD2.11	TC1.1.28.1 TC1.1.28.2
VAP1.33.1	Upon reception of the HW_COM.PTA.SystemCountersResetRequest, the DCM application shall reset MemoryNonVolatile.SystemCountersFile and update the counters related elements and then send HW_COM.DMS.SystemCountersReset Response	S3.2.7.24		SFD1.1.9.6.4		SD5.24 SD5.34	TC1.1.32.2
VAP1.34.1	Upon reception of HW_COM.PTA.SystemCountersRequest, the DCM application shall send HW_COM.DMS.SystemCountersResponse containing Memory NonVolatile.SystemCountersFile.	S3.2.7.25		SFD1.1.9.6.4		SD5.7 SD5.8 SD5.11 SD5.12 SD5.19 SD5.20	TC1.1.32.1
VAP1.35.1	Upon reception of HW_COM.PTA.MemoryDumpRequest from the PTA, the DCM application shall send HW_COM.DMS.MemoryDumpResponse containing MemoryVolatile.Content according to requested range.	S3.2.7.26				SD5.7 SD5.8	TC1.1.26.1
VAP1.36.1.1	The DCU Application shall use less than 90% of Volatile memory.	S3.3.1				Note 1	TC1.1.36.2
VAP1.36.1.2	The DCU Application shall use less than 45% of the Non-Volatile memory.	S3.3.1				Note 1	TC1.1.36.2
VAP1.36.2.1	DCM application shall use less than 50% of available processing time.	S3.3.2				Note 1	TC1.1.37.2
VAP1.36.3.1.1	The MemoryNonVolatile.EventLogFile shall contain 1500 events	S3.4.1				SD3.4	TC1.1.11.3
VAP1.36.3.1.2	The MemoryNonVolatile.EventLogFile shall be a circular buffer (after the 1500th event is logged, the 1501st shall overwrite the first entry)	S3.4.1				SD3.4	TC1.1.11.4
VAP1.36.3.1.3	Each event in the MemoryNonVolatile.EventLogFile shall contains the following data : • Timestamp • Event code (event number) • 16 attributes of each event (bit field) • System status (for timestamp validity) • Integrity check result for each event	S3.4.1				SD3.4	TC1.1.11.1
VAP1.36.3.2.1	The DCM application shall be able to capture SACS.Door.Snapshot. NbOfFiles snapshot files.	S3.4.2				SD7.3	TC1.1.11.5
VAP1.36.3.2.2	The DCM application shall be able to store up to SACS.Door.Snapshot.NbOfFiles snapshot files when its panel is mechanically cut-out	S3.4.2				SD7.3	TC1.1.21.1
VAP1.36.3.2.3	Each snapshot file shall contain (SACS.Door.Snapshot.Capacity) samples of the operating parameters status. Sample shall be taken every 1000/SACS.Door.Snapshot.Period milliseconds: SACS.Door.Snapshot.Capacity/2 samples taken before the event, 1 at event time, and SACS.Door.Snapshot.Capacity/2 after the event occurs	S3.4.2				SD7.3	TC1.1.11.5
VAP1.36.3.2.4	Each MemoryNonVolatile.SnapshotFile stored in DCM shall contain the data displayed in Table 3-25.	S3.4.2				SD7.3	TC1.1.11.5
VAP1.36.3.3.1	The MemoryNonVolatile.OperatingParametersDescriptionFile shall define the parameters that are retrievable from the PTA	S3.4.3				Note 1	Note 1
VAP1.36.3.3.2	These parameters shall be retrieved through the use of snapshot or RTDM	S3.4.3				SD7.3	Note 1

SRS Reqmt. ID	Requirements Description (From SRS)	SRS Ref. Section	TS Tag & Section	SFD / System Ref. ID	SDD Specific Ref. ID	SDD Generic Ref. ID	SITP Ref. ID
VAP1.36.3.3.3	The file shall contain records displayed in Table 3-27	S3.4.3				SD7.3	Note 1
VAP1.36.3.4.1	The MemoryNonVolatile.EventDescriptionFile shall be composed of a header and a list of records. The EDF shall be part of the DCM application code and shall not be changeable without upgrading the application	S3.4.4				Note 1	Note 1
VAP1.36.3.4.2	The event codes assignation in the EDF shall begin with “1” and be declared in ascending order up to a maximum of 1999	S3.4.4				Note 1	Note 1
VAP1.36.3.4.3	In order to maintain backward version compatibility, no event code shall be deleted or replaced	S3.4.4				Note 1	Note 1
VAP1.36.3.4.4	Events that are no longer used in newer software versions will remain in the list, and new events are added to the end of the existing list	S3.4.4				Note 1	Note 1
VAP1.36.3.4.5	The version of the EDF shall be incremented every time the EDF is updated in a released DCM application	S3.4.4				Note 1	Note 1
VAP1.36.3.4.6	The EDF shall contain records displayed in Table 3-28	S3.4.4				Note 1	Note 1
VAP1.36.3.5.1	The Configuration File shall contain adjustable door parameters.	S3.4.5				Note 1	TC1.1.28.1 TC1.1.28.2
VAP1.36.3.5.2	The Configuration File format shall contain the following data : • Variables formatted as per Appendix A • Integrity check result for the Configuration File	S3.4.5				Note 1	TC1.1.28.1 TC1.1.28.2
VAP1.36.3.6.1	There shall be one pair of MemoryNonVolatile.DoorMovementinformation Files ready for transfer to the PTA, empty if needed, at any time. The first one for the last opening sequence and the second one file for the last closing sequence	S3.4.6				Note 1	TC1.1.11.6
VAP1.36.3.6.2	The door movement information file shall contain the data displayed in Table 3-29.	S3.4.6				Note 1	TC1.1.11.7
VAP1.36.3.7.1	The system counters files shall store permanent counters of the door cycles, running time and the event occurrences in MemoryNonVolatile, using the format displayed in Table 3-30.	S3.4.7				Note 1	Note 1
VAP1.38.1	DCM application shall remain in a safe and predictable condition while initializing and be fully functional less than 60 seconds after a power-up or reset and stay functional as long as it is powered.	S3.6.2				Note 1	TC1.1.37.1
VAP1.38.2	To ensure availability, system must run under 1-second watchdog supervision; a failure to reset the watchdog timer within the pre-defined time shall trigger the re-initialization.	S3.6.2				SD1.4	Note 1

Note 1: Requirement traceability link will be added during the “SCT-600 Software Coding and Testing” phase.

3.3 PTU Application

Table 3-3 provides a central point of references for the software requirements. It provides a starting point for an upward and downward traceability of all allocated software requirements.

The table refers to the following:

- SRS - Generic Portable Test Application, Rev. 1, P/N 201111274
  - SRS Requirement ID: Rational RequisitePro Requirement Tag number
  - Requirements Description: SRS requirement text
  - SRS Reference Section: Section number from the SRS document.
- Rotem Technical Specification and Section
  - RTS: Rational RequisitePro Specification Requirement Tag number for Rotem Technical Specification
  - SO Section: Section of the System Overview for Side Door System Denver EMU P/N REDE113762
  - PTS Section: Section of the Procurement Technical Specification of Entrance Door System P/N REDE112784
- SFD – Door System, Rev. 4, P/N 201111185
  - SFD / System Reference ID: Rational RequisitePro Specification Requirement Tag number
- SDD - Generic Portable Test Application, Rev. 1, P/N 201111295
  - SDD Generic Reference ID: Rational RequisitePro Design Requirement Tag number
- SITP- Portable Test Unit Application, Rev. 1, P/N 201111464
  - SITP Reference ID: Rational RequisitePro Test Requirement Tag number

Table 3-3: PTU Application Software Requirements Traceability Matrix

SRS Reqmt ID	Requirements Description (From SRS)	SRS Ref. Section	TS Tag & Section	SFD / System Ref. ID	SDD Ref. ID	SITP Ref. ID
VAP3.1	GUI	S3.1.4			SD11.3	TC3.34.1
VAP3.2.1	Before sending HW_COM.PTA.Message, the PTA shall include the integrity check result and the Controller ID value of the CPU it want to address into the message.	S3.2.1.1			SD11.4.2.1	TC3.3.3 TC3.3.9 TC3.4.1
VAP3.2.2	Upon reception of HW_COM.DMS.Message or HW_COM.BootLoader. Message, the PTA shall verify the integrity of the message content and the validity of the contained command. If found valid, the PTA shall execute the command from the incoming message.	S3.2.1.1			SD11.4.2.1	TC3.3.3 TC3.3.9 TC3.4.1
VAP3.3.1	The PTA shall be able to log into CPU n°1 to n°X application at the same time.	S3.2.1.2		SFD1.3.1	SD11.3.1 SD11.4.1.3 SD11.4.1.5	TC3.3.2 TC3.3.8
VAP3.3.2	When the PTA is started, the PTA shall be able to detect if the DMS is present	S3.2.1.2		SFD1.3.1	SD11.4.1.1 SD11.4.1.5	TC3.3.1
VAP3.3.3	The PTA shall be able to login to the DMS Application and the DMS Boot Loader	S3.2.1.2		SFD1.3.1	SD11.4.1.5	TC3.3.2 TC3.3.8
VAP3.3.4	Upon reception of GUI.User.LoginRequest from the User, the PTA shall send HW_COM.PTA.LoginRequest and receive HW_COM.DMS.LoginResponse or HW_COM. BootLoader.LoginResponse in return. The PTA shall then send GUI.PTA.LoginResponse to the User.	S3.2.1.2		SFD1.3.1	SD11.4.1.5	TC3.3.2 TC3.3.8

SRS Reqmt ID	Requirements Description (From SRS)	SRS Ref. Section	TS Tag & Section	SFD / System Ref. ID	SDD Ref. ID	SITP Ref. ID
VAP3.4.1	Immediately after start-up, the PTA shall verify if an connection with a DMS is possible and try to log into that DMS by sending HW_COM.PTA.LoginRequest and receive HW_COM.DMS.LoginResponse or HW_COM.BootLoader.LoginResponse in return.	S3.2.1.3			SD11.4.1.1	TC3.3.1
VAP3.5.1	Upon reception of GUI.User.LogoutRequest from the User, the PTA shall send HW_COM.PTA.LogoutRequest and receive HW_COM.DMS.LogoutResponse or HW_COM.BootLoader.LogoutResponse in return. The PTA shall then send GUI.PTA.LogoutResponse to the User.	S3.2.1.4		SFD1.3.1	SD11.4.1.5	TC3.3.4 TC3.3.10
VAP3.5.2	If the communication is accidentally interrupted, the PTA shall be able to detect the communication error and logout.	S3.2.1.4		SFD1.3.1	SD11.4.1.5	TC3.3.5 TC3.3.11
VAP3.5.3	When the PTA is closed by the user, the PTA notify the DMS to disconnect automatically.	S3.2.1.4		SFD1.3.1	SD11.4.1.5	TC3.3.7 TC3.3.13
VAP3.6.1	Upon reception of GUI.User.RebootRequest from the User, the PTA shall send HW_COM.PTA.RebootRequest. The PTA shall then send GUI.PTA.RebootResponse to the User.	S3.2.1.5			SD11.4.3.1	TC3.3.15
VAP3.7.1	Every 30 seconds, the PTA shall send HW_COM.PTA.HeartbeatRequest to the Boot Loader or DMS and receive respectively HW_COM.BootLoader.HeartbeatResponse or HW_COM.DMS.HeartbeatResponse in answer, to ensure that a working connection is still established between the Boot Loader/DMS and the PTA.	S3.2.1.6			SD11.4.1.5	TC3.3.6 TC3.3.12
VAP3.8.1	Upon reception of GUI.User.EventStatusRequest from the User, the PTA shall send HW_COM.PTA.EventStatusRequest and receive HW_COM.DMS. EventStatusResponse in return. The PTA shall then send GUI.PTA.EventStatusResponse to the User.	S3.2.1.7		SFD1.3.5	SD11.3.10 SD11.4.5.3	TC3.11.3
VAP3.8.2	The PTA shall display the status for CPU n°1 to CPU n°X DMS applications.	S3.2.1.7		SFD1.3.5	SD11.3.10 SD11.4.5.3	TC3.11.3
VAP3.8.3	When the user opens the DMS status monitor; the PTA software shall keep the status updated in real-time.	S3.2.1.7		SFD1.3.5	SD11.3.10 SD11.4.5.3	TC3.11.1 TC3.11.2
VAP3.8.4	When the user closes the DMS status monitor, the PTA shall stop updating the status	S3.2.1.7		SFD1.3.5	SD11.3.10 SD11.4.5.3	TC3.9.3
VAP3.8.5	The DMS status shall include the following information of DMS if applicable to the system: door state, power supply, DMS inputs, DMS outputs, MDC.	S3.2.1.7		SFD1.3.5	SD11.3.10 SD11.4.5.3	TC3.11.2 TC3.11.4
VAP3.8.6	System status and functional information shall be provided in a graphical format, superimposed over a system schematic or diagrammatic. When a graphic presentation is not practical, system status and functional information shall be provided in english language statements and relevant engineering units, and shall be formatted for simple readability.	S3.2.1.7		SFD1.3.5	SD11.3.10 SD11.4.5.3	TC3.11.4 TC3.11.5
VAP3.9.1	The PTA shall display operating parameters for CPU n°1 to CPU n°X DMS Applications.	S3.2.1.8			SD11.3.9 SD11.4.5.3	TC3.9.1 TC3.9.2
VAP3.9.2	Upon reception of GUI.User.MonitoringRequest from user, the PTA shall send HW_COM.PTA.MonitoringRequest to the DMS and receive HW_COM.DMS. MonitoringResponse in response, and then send GUI.PTA.MonitoringResponse to the user.	S3.2.1.8			SD11.3.9 SD11.4.5.3	TC3.9.1 TC3.9.2
VAP3.9.3	The software shall keep updating the real-time data until the User stop requesting the real-time monitoring.	S3.2.1.8			SD11.3.9 SD11.4.5.3	TC3.9.3
VAP3.10.1	The PTA shall provide the option to download the log file from the CPU n°1 or CPU n°X DMS application.	S3.2.1.9		SFD1.3.2	SD11.4.4.4	TC3.22.1
VAP3.10.2	Upon GUI.User.EventLogFileRequest, the PTA software shall send HW_COM.PTA.EventLogFileRequest to the DMS, receive HW_COM.DMS. EventLogFileResponse in response and save the event log it contains into Memory.EventLog. The PTA shall then send GUI.PTA.EventLogFileResponse to the User.	S3.2.1.9		SFD1.3.2	SD11.4.4.4	TC3.22.1 TC3.22.2 TC3.22.3
VAP3.10.3	When a download error occurs, the error information shall be displayed.	S3.2.1.9		SFD1.3.2	SD11.4.4.4	TC3.22.7
VAP3.10.4	Download progression shall appear during event log file transfer.	S3.2.1.9		SFD1.3.2	SD11.3.6 SD11.4.2.3 SD11.4.4.4	TC3.22.1
VAP3.11.1	When GUI.User.EventLogFileDisplayRequest is received by the PTA, it shall send GUI.PTA.EventLogFileDisplay to the User, displaying the event log in the sequence order.	S3.2.1.10		SFD1.3.2	SD11.3.16 SD11.4.4.4	TC3.22.3

SRS Reqmt ID	Requirements Description (From SRS)	SRS Ref. Section	TS Tag & Section	SFD / System Ref. ID	SDD Ref. ID	SITP Ref. ID
VAP3.11.2	The Log Table shall present the decoded Memory.EventLog database to the user and shall allow sorting by any of its fields.	S3.2.1.10		SFD1.3.2	SD11.3.16 SD11.4.4.4	TC3.22.3 TC3.22.4
VAP3.12.1	The PTA shall provide the option to delete the log file of the CPU n°1 or CPU n°X DMS application.	S3.2.1.11			SD11.3.15 SD11.3.26 SD11.4.4.5	TC3.22.8
VAP3.12.2	Upon reception of GUI.User.EraseLogFileRequest from User, the PTA shall send GUI.PTA.ConfirmationRequest to the User, and get GUI.User. ConfirmationResponse in return. The PTA shall then send HW_COM.PTA. EraseLogFileRequest to the DMS and receive HW_COM.DMS.EraseLogFileResponse when the DMS deleted the event log. The PTA shall then send GUI.PTA. EraseLogFileResponse to the User.	S3.2.1.11			SD11.4.4.5	TC3.22.8 TC3.22.9 TC3.22.10
VAP3.13.1	The PTA shall provide the option to delete the log file of the CommunicationGateway application of any DMS. **(Not applicable to the Denver Project)**	S3.2.1.12		N/A	N/A	N/A
VAP3.13.2	Upon reception GUI.User.DeleteCommunicationGatewayLogRequest from the User, the PTA shall send GUI.PTA.ConfirmationRequest for confirmation and receive GUI.User.ConfirmationResponse in return. The PTA shall then send HW_COM.PTA. DeleteCommunicationGatewayLogRequest to the DMS and receive HW_COM.DMS. DeleteCommunicationGatewayLogResponse upon deletion of the CommunicationGateway log. The PTA shall then send GUI.PTA. DeleteCommunicationGatewayLogResponse to the user to notify the success of the deletion. (**Not applicable to the Denver Project)**	S3.2.1.12		N/A	N/A	N/A
VAP3.14.1	The PTA shall provide the option to download the RTDM snapshot file from CPU n°1 or CPU n°X DMS application.	S3.2.1.13		SFD1.3.3	SD11.3.19 SD11.4.5.4	TC3.22.11
VAP3.14.2	Upon reception of GUI.User.SnapshotFileRequest from user, the PTA shall send HW_COM.PTA.SnapshotFileRequest, receive HW_COM.DMS. SnapshotFileResponse in return and save its content into Memory.Snapshot. The PTA shall then send GUI.PTA.SnapshotFileResponse to the user.	S3.2.1.13		SFD1.3.3	SD11.4.5.4	TC3.22.11 TC3.22.12 TC3.22.13
VAP3.14.3	Upon reception, Memory.Snapshot shall be uncompressed, decoded according to OPDF data format for each operating parameter and added to snapshot list. The PTA software tool shall be able to view all snapshot contents.	S3.2.1.13		SFD1.3.3	SD11.3.19 SD11.3.20 SD11.4.5.4	TC3.22.13
VAP3.14.4	When download error occurs, the error information shall be displayed to the user.	S3.2.1.13		SFD1.3.3	SD11.4.5.4	TC3.22.14
VAP3.14.5	Download progression shall appear during snapshot file transfer.	S3.2.1.13		SFD1.3.3	SD11.3.6 SD11.4.2.3 SD11.4.5.4	TC3.22.12
VAP3.15.1	The PTA shall provide the option to delete snapshot file to CPU n°01 or CPU n°X DMS application.	S3.2.1.14			SD11.3.18 SD11.4.5.5	TC3.22.15
VAP3.15.2	Upon reception of GUI.User.EraseSnapshotFileRequest from the User, the PTA shall send GUI.PTA.ConfirmationRequest for confirmation, and receive GUI.User.ConfirmationResponse in return. The PTA shall then send HW_COM.PTA. EraseSnapshotFileRequest to the DMS and receive HW_COM.DMS. EraseSnapshotFileResponse after the snapshot file has been deleted. The PTA shall then send GUI.PTA.EraseSnapshotFileResponse to notify the User of the success of the operation	S3.2.1.14			SD11.4.5.5	TC3.22.15 TC3.22.16 TC3.22.17
VAP3.16.1	The PTA shall be able to request self test results to CPU n°1 to CPU n°X DMS applications	S3.2.1.15		SFD1.3.6	SD11.3.22 SD11.4.6.1	TC3.10.1
VAP3.16.2	Upon reception of GUI.User.SelfTestRequest from the User, , the PTA shall send HW_COM.PTA.SelfTestRequest to the DMS and receive HW_COM.DMS. SelfTestResponse containing the self-test results. The PTA shall then send GUI.PTA. SelfTestResponse to the User, displaying the results on the screen.	S3.2.1.15		SFD1.3.6	SD11.4.6.1	TC3.10.1 TC3.10.2
VAP3.17.1	The PTA shall provide the option to upgrade CPU n°1 or CPU n°X DMS application.	S3.2.1.16		SFD1.3.10	SD11.3.7 SD11.3.8 SD11.4.3.1	TC3.24.1

SRS Reqmt ID	Requirements Description (From SRS)	SRS Ref. Section	TS Tag & Section	SFD / System Ref. ID	SDD Ref. ID	SITP Ref. ID
VAP3.17.2	Upon reception of GUI.User.DMSApplicationUpgradeRequest from the User, the PTA shall send GUI.PTA.ConfirmationRequest for confirmation and receive GUI.User.ConfirmationResponse in answer. The PTA shall then send HW_COM.PTA.DMSApplicationUpgradeRequest, containing the application code, to the Boot Loader and receive HW_COM.BootLoader.DMSApplicationUpgradeResponse upon completion of the application upgrade. The PTA shall then send GUI.PTA.DMSApplicationUpgradeResponse to notify the User of the success of the operation.	S3.2.1.16		SFD1.3.10	SD11.4.3.1	TC3.24.1 TC3.24.2
VAP3.17.3	Errors compromising the integrity of the DMS software shall interrupt the DMS upgrade and be reported to the user.	S3.2.1.16		SFD1.3.10	SD11.4.3.1	TC3.24.4 TC3.24.5 TC3.24.6
VAP3.17.4	Download progression shall appear during the DMS application file transfer.	S3.2.1.16		SFD1.3.10	SD11.3.6 SD11.4.2.3 SD11.4.3.1	TC3.24.3
VAP3.18.1	The PTA shall perform upgrade of the Boot Loader application saved as Memory.BootLoaderApplicationFile through the HW_COM interface. The implementation of this function is specific to the CPU being used in the DMS and shall be define in the design phase.	S3.2.1.17		SFD1.3.10	SD11.4.2.4	TC3.25.1
VAP3.19.1	The PTA shall provide the option to configure CPU n°1 or CPU n°X DMS application.	S3.2.1.18	RTS1.2.1.1.1 PTS Section 5: Re-open Distance PTU Programmable RTS1.2.1.1.2 PTS Section 5: Delay before Re-closing PTU Programmable RTS1.2.1.1.3 PTS Section 5: Obstruction Re-tries PTU Programmable RTS1.2.1.1.4 PTS Section 5: Obstruction Count Reset	SFD1.3.14	SD11.4.9.1	TC3.12.1 TC3.13.1 TC3.14.1 TC3.15.1 TC3.16.1 TC3.17.1 TC3.18.1
VAP3.19.2	Upon reception of GUI.User.ReadDMSConfigurationRequest from the User, the PTA shall send HW_COM.PTA.ReadDMSConfigurationRequest to the DMS and receive GUI.PTA.ReadDMSConfigurationResponse. The PTA shall then send GUI.PTA.ReadDMSConfigurationResponse to the User, displaying the current parameters.	S3.2.1.18	RTS1.2.1.1.1 PTS Section 5: Re-open Distance PTU Programmable RTS1.2.1.1.2 PTS Section 5: Delay before Re-closing PTU Programmable RTS1.2.1.1.3 PTS Section 5: Obstruction Re-tries PTU Programmable RTS1.2.1.1.4 PTS Section 5: Obstruction Count Reset	SFD1.3.14	SD11.4.9.1	TC3.12.1 TC3.13.1 TC3.14.1 TC3.15.1 TC3.16.1 TC3.17.1 TC3.18.1

SRS Reqmt ID	Requirements Description (From SRS)	SRS Ref. Section	TS Tag & Section	SFD / System Ref. ID	SDD Ref. ID	SITP Ref. ID
VAP3.20.1	The PTA shall provide the option to configure CPU n°1 or CPU n°X DMS application.	S3.2.1.19	RTS1.2.1.1.1 PTS Section 5: Re-open Distance PTU Programmable RTS1.2.1.1.2 PTS Section 5: Delay before Re-closing PTU Programmable RTS1.2.1.1.3 PTS Section 5: Obstruction Re-tries PTU Programmable RTS1.2.1.1.4 PTS Section 5: Obstruction Count Reset	SFD1.3.14	SD11.3.25 SD11.4.9.2	TC3.12.3 TC3.13.3 TC3.14.3 TC3.15.3 TC3.16.2 TC3.17.3 TC3.18.3
VAP3.20.2	Upon reception of GUI.User.WriteDMSConfigurationRequest from the User, the PTA shall send HW_COM.PTA.WriteDMSConfigurationRequest to the DMS and receive GUI.PTA.WriteDMSConfigurationResponse upon modification of the parameters by the DMS. The PTA shall then send GUI.PTA.WriteDMSConfigurationResponse to the User..	S3.2.1.19	RTS1.2.1.1.1 PTS Section 5: Re-open Distance PTU Programmable RTS1.2.1.1.2 PTS Section 5: Delay before Re-closing PTU Programmable RTS1.2.1.1.3 PTS Section 5: Obstruction Re-tries PTU Programmable RTS1.2.1.1.4 PTS Section 5: Obstruction Count Reset	SFD1.3.14	SD11.4.9.2	TC3.12.2 TC3.12.3 TC3.12.4 TC3.13.2 TC3.13.3 TC3.13.4 TC3.14.2 TC3.14.3 TC3.14.4 TC3.15.2 TC3.15.3 TC3.15.4 TC3.16.2 TC3.16.3 TC3.16.4 TC3.16.5 TC3.17.2 TC3.17.3 TC3.18.2 TC3.18.3 TC3.18.4
VAP3.21.1	The PTA shall be able to detect error occurred in communication, such as packet integrity error and time out, when receiving HW_COM.DMS.Message or HW_COM. BootLoader.Message or when sending HW_COM.PTA.Message.	S3.2.1.20			SD11.4.2.1 SD11.4.2.2 SD11.4.2.3	TC3.4.1
VAP3.21.2	A mechanism shall be implemented to avoid non-end communication waiting.	S3.2.1.20			SD11.4.2.1 SD11.4.2.2 SD11.4.2.3	TC3.4.2
VAP3.21.3	If PTA fails to recover from a communication error, the error information shall be reported to the user through GUI.PTA.ErrorReportDisplay.	S3.2.1.20			SD11.4.2.1 SD11.4.2.2 SD11.4.2.3	TC3.4.2 TC3.4.3
VAP3.22.1	The PTA shall provide the option to retrieve the time from CPU n°1 to n°X DMS application.	S3.2.1.21		SFD1.3.7	SD11.4.10.1	TC3.19.1 TC3.19.2
VAP3.22.2	Upon reception of GUI.User.LocalTimeRequest, the PTA shall send HW_COM.PTA.LocalTimeRequest to the DMS and receive HW_COM.DMS. LocalTimeResponse in return, containing the time of the CPU. The PTA shall then send GUI.PTA.LocalTimeResponse to the User, displaying the DMS time.	S3.2.1.21		SFD1.3.7	SD11.4.10.1	TC3.19.1 TC3.19.2
VAP3.22.3	The DMS date (mm/dd/yyyy) and time (hh:mm:ss AM/PM) shall be displayed in local time.	S3.2.1.21		SFD1.3.7	SD11.3.23	TC3.19.2
VAP3.23.1	The PTA shall provide the option to set the time from CPU n°1 to n°x DMS application.	S3.2.1.22			SD11.4.10.2	TC3.19.1

SRS Reqmt ID	Requirements Description (From SRS)	SRS Ref. Section	TS Tag & Section	SFD / System Ref. ID	SDD Ref. ID	SITP Ref. ID
VAP3.23.2	Upon reception of GUI.User.SetTimeRequest, the PTA shall send HW_COM.PTA. SetTimeRequest to the DMS and receive HW_COM.DMS. SetTimeResponse in return, containing the time of the CPU. The PTA shall then send GUI.PTA. SetTimeResponse to the User, displaying the DMS time.	S3.2.1.22			SD11.4.10.2	TC3.19.1 TC3.19.2
VAP3.24.1	The PTA shall provide the option to retrieve the software version from CPU n°1 or CPU n°X application.	S3.2.1.23		SFD1.3.9	SD11.3.5	TC3.8.1
VAP3.24.2	After login, the PTA shall send HW_COM.PTA.SoftwareVersionsRequest to the DMS and receive HW_COM.DMS.SoftwareVersionsResponse or HW_COM.BootLoader. SoftwareVersionsResponse in answer, containing the DMS SCI items software versions.	S3.2.1.23		SFD1.3.9	SD11.4.11.1	TC3.8.1 TC3.8.2
VAP3.24.3	When available, the DMS software versions shall be displayed	S3.2.1.23		SFD1.3.9	SD11.3.5	TC3.8.3
VAP3.24.4	The DMS software versions shall include the DMS application version, the Boot Loader version and , if applicable, CommunicationGateway application version.	S3.2.1.23		SFD1.3.9	SD11.3.5 SD11.4.11.1	TC3.8.3
VAP3.25.1	The PTA shall provide the option to retrieve the Location identification and car number from CPU n°1 or CPU n°X application.	S3.2.1.24		SFD1.3.8	SD11.4.1.6	TC3.7.1
VAP3.25.2	Automatically after login into the DMS, the PTA shall send HW_COM.PTA.LocationIdentificationRequest and HW_COM.PTA.CarNumberRequest to the DMS and receive respectively HW_COM.DMS.LocationIdentificationResponse and HW_COM.DMS.CarNumberResponse from the DMS, containing respectively the location identification and the car number.	S3.2.1.24		SFD1.3.8	SD11.4.1.6	TC3.7.1 TC3.7.2
VAP3.25.3	Upon reception of the information, the PTA shall send GUI.PTA. DMSLocationIdentificationDisplay to the User, displaying the door location identification.	S3.2.1.24		SFD1.3.8	SD11.3.2	TC3.7.3
VAP3.25.4	Upon reception of the information, the PTA shall send GUI.PTA. DMSCarNumberDisplay to the User, displaying the door car number.	S3.2.1.24		SFD1.3.8	SD11.3.2 SD11.4.1.6	TC3.7.4
VAP3.26.1	The PTA shall provide the option to retrieve the code status from CPU n°1 or CPU n°X application.	S3.2.1.25			SD11.3.24 SD11.4.11.3	TC3.6.1 TC3.6.2 TC3.6.3
VAP3.26.2	Upon reception of GUI.User.CodeStatusRequest from the User, the PTA shall send HW_COM.PTA.CodeStatusRequest to the DMS and receive HW_COM.DMS. CodeStatusResponse or HW_COM.BootLoader.CodeStatusResponse in return, containing the DMS code status. The PTA shall then send GUI.PTA. CodeStatusResponse to the User, displaying the DMS code status on screen.	S3.2.1.25			SD11.4.11.3	TC3.6.1 TC3.6.2 TC3.6.3
VAP3.26.3	The DMS code status shall include the DMS application, the DMS Boot Loader application, and, if applicable, the CommunicationGateway application code status.	S3.2.1.25			SD11.4.11.3	TC3.6.1 TC3.6.2 TC3.6.3
VAP3.27.1	The PTA shall provide the option to retrieve the volatile memory content from the CPU n°1 or CPU n°X application.	S3.2.1.26			SD11.3.17 SD11.4.7.1	TC3.20.1
VAP3.27.2	The PTA shall provide the option to set the starting address and the end address of the memory zone to download	S3.2.1.26			SD11.3.17 SD11.4.7.1	TC3.20.3
VAP3.27.3	Upon reception of GUI.User.MemoryDumpRequest from the User, the PTA shall send HW_COM.PTA.MemoryDumpRequest to the DMS, receive HW_COM.DMS .MemoryDumpResponse in return and save its content into Memory.DMSMemoryDump . The PTA shall then send GUI.PTA.MemoryDumpResponse to the User, displaying the memory dump result on screen.	S3.2.1.26			SD11.4.7.1	TC3.20.1 TC3.20.2
VAP3.27.4	The DMS volatile memory contents shall be displayed in both Hex and ASCII format.	S3.2.1.26			SD11.3.17 SD11.4.7.1	TC3.20.1 TC3.20.2
VAP3.27.5	Download progression shall appear during DMS memory file transfer.	S3.2.1.26			SD11.4.2.3 SD11.4.7.1	TC3.20.1
VAP3.28.1	The PTA shall provide the option to download the door movement information file from CPU n°1 and CPU n°X DMS application.	S3.2.1.27		SFD1.3.15	SD11.3.14 SD11.4.8.1	TC3.21.1

SRS Reqmt ID	Requirements Description (From SRS)	SRS Ref. Section	TS Tag & Section	SFD / System Ref. ID	SDD Ref. ID	SITP Ref. ID
VAP3.28.2	Upon reception of GUI.User.DoorMovementInformationFileRequest from the User, the PTA shall send HW_COM.PTA.DoorMovementInformationFileRequest to the DMS, receive HW_COM.DMS.DoorMovementInformationFileResponse in return and save its content into Memory.DoorMovementInformation. The PTA shall then send GUI.PTA.DoorMovementInformationFileResponse to the User, displaying the data on screen.	S3.2.1.27		SFD1.3.15	SD11.4.8.1	TC3.21.1 TC3.21.2 TC3.21.3
VAP3.28.3	The PTA shall present door movement information graphically to the user.	S3.2.1.27		SFD1.3.15	SD11.3.14 SD11.4.8.1	TC3.21.3
VAP3.28.4	When a download error occurs the Error Information shall be displayed.	S3.2.1.27		SFD1.3.15	SD11.4.8.1	TC3.21.4
VAP3.28.5	Download progression shall appear during graph file transfer.	S3.2.1.27		SFD1.3.15	SD11.3.6 SD11.4.2.3 SD11.4.8.1	TC3.21.1
VAP3.29.1	Upon reception of GUI.User.EventDescriptionFileRequest from the User, the PTA shall send HW_COM.PTA.EventDescriptionFileRequest to the DMS, receive HW_COM.DMS. EventDescriptionFileResponse in return and save its content into Memory. EventDescriptionFile. The PTA shall then send GUI.PTA.EventDescriptionFileResponse to the User, displaying the data on screen.	S3.2.1.28			SD11.4.4.1	TC3.5.1 TC3.5.2
VAP3.29.2	The DMS EDF Contents shall be able to be browsed by the user.	S3.2.1.28			SD11.3.11 SD11.3.12 SD11.4.4.1	TC3.5.1
VAP3.29.3	Download progression shall appear during EDF file transfer.	S3.2.1.28			SD11.3.6 SD11.4.4.1	TC3.5.1
VAP3.30.1	Upon reception of GUI.User.OperatingParametersDescriptionFileRequest from the User, the PTA shall send HW_COM.PTA.OperatingParametersDescriptionFileRequest to the DMS, receive HW_COM.DMS.OperatingParametersDescriptionFileResponse in return and save its content into Memory.OperatingParameterDescriptionFile. The PTA shall then send GUI.PTA.OperatingParametersDescriptionFileResponse to the User, displaying the data on screen.	S3.2.1.29			SD11.4.5.1	TC3.5.3 TC3.5.4
VAP3.30.2	The DMS OPDF Contents shall be able to be browsed by the user.	S3.2.1.29			SD11.3.13 SD11.4.5.1	TC3.5.3
VAP3.30.3	Download progression shall appear during OPDF file transfer.	S3.2.1.29			SD11.3.6 SD11.4.2.3 SD11.4.5.1	TC3.5.3
VAP3.31.1	Upon reception of GUI.User.SystemCountersRequest from the User, the PTA shall send HW_COM.PTA.SystemCountersRequest to the DMS, receive HW_COM.DMS. SystemCountersResponse in return and save its content into Memory.SystemCounters. The PTA shall then send GUI.PTA.SystemCountersResponse to the User, displaying the data on screen.	S3.2.1.30		SFD1.3.4	SD11.4.12.1	TC3.30.1 TC3.30.2
VAP3.31.2	The DMS System Counters Contents shall be displayed to the user.	S3.2.1.30		SFD1.3.4	SD11.3.21 SD11.4.12.1	TC3.30.1
VAP3.31.3	Download progression shall appear during System counters file transfer. The access to other PTA functions should be disabled until the data transfer is completed or cancelled.	S3.2.1.30		SFD1.3.4	SD11.3.6 SD11.4.2.3	TC3.30.1
VAP3.32.1	Upon reception of GUI.User.SystemCountersResetRequest from the User, the PTA shall send GUI.PTA.ConfirmationRequest for confirmation and receive GUI.User.ConfirmationResponse in return from the User. the PTA shall then send HW_COM.PTA.SystemCountersResetRequest to the DMS and receive HW_COM.DMS.SystemCountersResetResponse in return. The PTA shall then send GUI.PTA.SystemCountersResetResponse to the User, notifying him of the success of the reset	S3.2.1.31		SFD1.3.4	SD11.4.12.2 SD11.4.12.3 SD11.4.12.4 SD11.4.12.5 SD11.4.12.6 SD11.4.12.7	TC3.30.3
VAP3.33.1	Upon reception of GUI.User.OpenFileRequest from the User, the PTA shall send GUI.PTA.OpenFileResponse to the user, open the selected file from Memory. DataFile and display it on screen.	S3.2.1.32		SFD1.3.11	SD11.4.14.2	TC3.23.2 TC3.23.3
VAP3.33.2	Bad file name, type and file data shall be detected and reported to the user.	S3.2.1.32		SFD1.3.11	SD11.4.14.2	TC3.23.5

SRS Reqmt ID	Requirements Description (From SRS)	SRS Ref. Section	TS Tag & Section	SFD / System Ref. ID	SDD Ref. ID	SITP Ref. ID
VAP3.34.1	Upon reception of GUI.User.SaveFileRequest from the User, the PTA shall send GUI.PTA.SaveFileResponse to the user and save the selected file from Memory.DataFile.	S3.2.1.33		SFD1.3.2 SFD1.3.3	SD11.4.14.1	TC3.23.1 TC3.23.4
VAP3.35.1	Upon reception of GUI.User.PrintRequest from the User, the PTA shall send GUI.PTA.PrintResponse to the User and print the selected file (Memory.DataFile) in predefined format.	S3.2.1.34		SFD1.3.2 SFD1.3.3	SD11.4.15.1 SD11.4.15.2 SD11.4.15.3	TC3.29.1
VAP3.36.1	The PTA shall provide interfaces to allow user to change current working directory, to set up Hardware Interface(s) and to define login options	S3.2.1.35			SD11.3.3 SD11.4.1.3	TC3.2.1 TC3.2.2 TC3.2.3 TC3.32.1 TC3.32.2
VAP3.36.2	Upon reception of GUI.User.ConfigurePTARequest from the User, the PTA shall send GUI.PTA.ConfigurePTAResponse and save the changes into Memory.PTASettings.	S3.2.1.35			SD11.4.1.1 SD11.4.1.3	TC3.2.2 TC3.2.3
VAP3.36.3	Each time the PTA starts, it shall read PTA Settings from the system registry and set up the PTA.	S3.2.1.35			SD11.4.1.1 SD11.4.1.3	TC3.2.1 TC3.2.2 TC3.2.3
VAP3.36.4	The setup error shall be reported to the user.	S3.2.1.35			SD11.4.1.1 SD11.4.1.3	TC3.2.4
VAP3.37.1	Upon reception of GUI.User.HelpRequest from the User, the PTA shall send GUI.PTA.HelpResponse to the User and start the Help File engine to load help into memory	S3.2.1.36		SFD1.3.12	SD11.4.1.7	TC3.28.1
VAP3.37.2	User shall be able to use this help engine to browse the help file.	S3.2.1.36		SFD1.3.12	SD11.4.1.7	TC3.28.1
VAP3.38.1	The PTA user shall be prohibited from performing the following requests unless the PTA detects the presence of the password with the proper Firm and User code: GUI.User.DMSApplicationUpgradeRequest, GUI.User.WriteDMSConfigurationRequest, GUI.User.EraseSnapshotFileRequest.	S3.2.1.37			SD11.3.3 SD11.4.1.4	TC3.27.1 TC3.27.2
VAP3.39.1	The time required to upload the entire software for the door system, including time to replace firmware embodied in CPUs, shall be no more than one hour.	S3.3			SD11.4.2.4 SD11.4.3	TC3.24.1 TC3.25.1
VAP3.41.1.1	The PTA installation shall make use of the Microsoft Windows Installer service, InstallShield or equivalent for Windows Installer for developing Setup procedures.	S3.7.1		SFD1.3.13	SD11.2	TC3.1.1 TC3.1.2 TC3.1.3
VAP3.41.1.2	The Uninstall program shall not require the user to insert a CD-ROM or any type of external media. It shall remove all files in the Program Files folder that it uniquely owns. The Uninstall program shall leave no remaining files or registry entries other than the files created by user.	S3.7.1		SFD1.3.13	SD11.2	TC3.31.2 TC3.31.3 TC3.31.4
VAP3.41.1.3	The Uninstall program should be registered so that it will be listed on the Uninstall page of the Add/Remove Programs utility included with Windows. Safeguards and protection that will prevent inadvertent deletions of programs must be provided as functions of the Uninstall procedure.	S3.7.1		SFD1.3.13	SD11.2	TC3.31.1

3.1 Neuron Gateway Application

Table 3-4 provides a central point of references for the software requirements. It provides a starting point for an upward and downward traceability of all allocated software requirements.

The table refers to the following:

- SRS – Neuron Gateway Application, Rev. 1, P/N 201111256
  - SRS Requirement ID: Rational RequisitePro Requirement Tag number
  - Requirements Description: SRS requirement text
  - SRS Reference Section: Section number from the SRS document.
- SFD – Door System, Rev. 4, P/N 201111185
  - SFD / System Reference ID: Rational RequisitePro Specification Requirement Tag number
- SDD – Neuron Gateway Application, Rev. 1, P/N 201111459
  - SDD Generic Reference ID: Rational RequisitePro Design Requirement Tag number
- SITP – Neuron Gateway Application, Rev. 1, P/N 201111460
  - SITP Reference ID: Rational RequisitePro Test Requirement Tag number

Table 3-4: Neuron Gateway Application Software Requirements Traceability Matrix

SRS Reqmt ID	Requirements Description (From SRS)	SRS Ref. Section	SFD / System Ref. ID	SDD Specific Ref. ID	SITP Ref. ID
VAP3.1.1	NGA shall set the Loc0:5 based on its physical location in the car (valid values: “DCU 1”, “DCU 2”, “DCU 3”, and “DCU 4”).	S3.2.1	SFD1.2	SD3.1	TC2.1.1 TC2.1.2 TC2.1.3 TC2.1.4
VAP3.1.2	NGA shall set the Domain0:5 value to zero, Subnet value to 1, DomainLength value to 0, and the node number based on its DCU_Location (DCU1~DCU4: 17 ~ 20).	S3.2.1		SD3.1	TC2.1.1 TC2.1.2 TC2.1.3 TC2.1.4
VAP3.1.3	NGA shall set the ProgID0:7 value to “DCU”.	S3.2.1		SD3.1	TC2.1.1 TC2.1.2 TC2.1.3 TC2.1.4
VAP3.1.4	NGA shall set the node self-document string to “DCU, Vapor, PPPP, DDDD” (where PPPP is supplier part number and DDDD is Neuron SCI revision date)	S3.2.1		SD3.1	TC2.1.1 TC2.1.2 TC2.1.3 TC2.1.4
VAP3.1.5	NGA shall set the Network Variable Index and the Sender/Receiver of all Network Variable Properties based on its DCU_Location.	S3.2.1		SD3.1	TC2.1.1 TC2.1.2 TC2.1.3 TC2.1.4
VAP3.2.1	Upon receiving the nvi00ObjRequest, NGA shall send object status:	S3.2.2		Note 2	TC2.9.1 TC2.9.2 TC2.9.3 TC2.9.4 TC2.9.5 TC2.9.6

SRS Reqmt ID	Requirements Description (From SRS)	SRS Ref. Section	SFD / System Ref. ID	SDD Specific Ref. ID	SITP Ref. ID
VAP3.2.1.1	Upon receiving RQ_NORMAL, NGA shall return the specified functional block to normal operation. (if the Node Object is specified then all functional blocks are applied.)	S3.2.2		Note 2	TC2.9.2
VAP3.2.1.2	Upon receiving RQ_UPDATE_STATUS, NGA shall update the SNVT_obj_status. (if the Node Object is specified then inclusive-OR of the status bits for all functional blocks are applied)	S3.2.2		Note 2	TC2.9.3
VAP3.2.1.3	Upon receiving RQ_REPORT_MASK, NGA shall set only used bits status to 1. (if the Node Object is specified then inclusive-OR of supported status bits for all functional blocks are applied.)	S3.2.2		Note 2	TC2.9.4
VAP3.2.1.4	Upon receiving RQ_CLEAR_STATUS, NGA shall clear the status bits of the specified functional block. (if the Node Object is specified then all functional blocks are applied.)	S3.2.2		Note 2	TC2.9.5
VAP3.2.2	Upon receiving the nvi00ObjRequest, NGA shall verify the object ID and the request:	S3.2.2		Note 2	TC2.9.6
VAP3.2.2.1	Only if the object ID is invalid, then NGA shall set the invalid_id bit to 1.	S3.2.2		Note 2	TC2.9.6
VAP3.2.2.2	Only if the request is invalid, then NGA shall set the invalid_request bit to 1.	S3.2.2		Note 2	TC2.9.6
VAP3.3.1	Upon receiving the nvi00TimeStamp from the car Network, the NGA software shall analyze the network time (year, month, day, hour, minute, second) for validity:	S3.2.3		SD3.11	TC2.3.1
VAP3.3.1.1	NGA shall verify the value of the year (valid value: 0, -1, 1990~2100);	S3.2.3		SD3.11	TC2.3.2 TC2.3.3 TC2.3.4 TC2.3.5 TC2.3.6
VAP3.3.1.2	NGA shall verify the value of the month (valid value: 0~12);	S3.2.3		SD3.11	TC2.3.2 TC2.3.3 TC2.3.4 TC2.3.5 TC2.3.6
VAP3.3.1.3	NGA shall verify the value of the day in the month (valid value: 0~31);	S3.2.3		SD3.11	TC2.3.2 TC2.3.3 TC2.3.4 TC2.3.5 TC2.3.6
VAP3.3.1.4	NGA shall verify the value of the hour (valid value: 0~23);	S3.2.3		SD3.11	TC2.3.2 TC2.3.3 TC2.3.4 TC2.3.5 TC2.3.6
VAP3.3.1.5	NGA shall verify the value of the minute (valid value: 0~59);	S3.2.3		SD3.11	TC2.3.2 TC2.3.3 TC2.3.4 TC2.3.5 TC2.3.6
VAP3.3.1.6	NGA shall verify the value of the second (valid value: 0~59).	S3.2.3		SD3.11	TC2.3.2 TC2.3.3 TC2.3.4 TC2.3.5 TC2.3.6
VAP3.3.2	If the Network Time Stamp is in valid range, NGA shall save it to the UTC_time. If invalid, the NGA shall store 1990-01-01 00h00.00.	S3.2.3		SD3.11	TC2.3.3 TC2.3.4 TC2.3.5 TC2.3.6
VAP3.3.3	Upon reception of the Time Stamp, NGA shall set the ReceivedTime (SPI_Frame.NGAtoDCU) = 1 for the duration of two Level-1 Protocol frame, and then return to 0. See also the SPI Media Access Function in section 3.2.8.	S3.2.3		SD3.11	TC2.3.7
VAP3.4.1	NGA software shall always set the number_of_files = 1, and the selected_file = index.	S3.2.4		SD3.12	TC2.7.1
VAP3.4.2	NGA software shall ignore the receive_timeout.	S3.2.4		SD3.12	TC2.7.1
VAP3.4.3	NGA shall accept only the file requests from the CDT (nvi00FileReq). For those requests that are not acceptable, NGA shall respond with status = FS_ACCESS_UNAVAIL.	S3.2.4		SD3.12	TC2.7.1

SRS Reqmt ID	Requirements Description (From SRS)	SRS Ref. Section	SFD / System Ref. ID	SDD Specific Ref. ID	SITP Ref. ID
VAP3.4.4	When NGA receives the FR_DIRECTORY_LOOKUP request with correct file index, it shall respond with the file status = FS_LOOKUP_OK. (if applicable)	S3.2.4		SD3.12	TC2.7.1
VAP3.4.5	When the NGA software receives the FR_OPEN_TO_SEND_RA, the FR_OPEN_TO_RECEIVE_RA, the FR_NUL, or an invalid request, then it shall respond with the status = FS_ACCESS_UNAVAIL.	S3.2.4		SD3.12	TC2.7.1
VAP3.4.6	When NGA receives the FR_OPEN_TO_RECEIVE request with correct file index, then it shall respond with the status = FS_XFER_UNDERWAY. (if applicable)	S3.2.4		SD3.12	TC2.7.1
VAP3.4.7	When NGA receives the FR_CLOSE_FILE or FR_CLOSE_DELETE_FILE request for any file index, it shall respond with the status = FS_XFER_OK and stop the transfer if in progress.	S3.2.4		SD3.12	TC2.7.1
VAP3.4.8	NGA shall use an Ftp_DataWindow size of 6 and a maximum packet size of 32 bytes.	S3.2.4		SD3.12	TC2.7.1
VAP3.4.9	When sending its Ftp_DataWindow packets with the Service type = Request/Response, NGA shall set the Transaction timer = tx_timer3:tx_timer0 and the retry3:retry0 = retry3:retry0.	S3.2.4		SD3.12	TC2.7.1
VAP3.4.10	NGA shall send its variable Length data window packets in Data0 … Data31, indicating its window number in Win3:0, and its packet number in Pkt2:0. See the Figure 3-3.	S3.2.4		SD3.12	TC2.7.1
VAP3.4.11	NGA shall accept responses from the CDT to its Request/Response Ftp_DataWindow packets in the NextPacketExpected. NGA shall either retransmit packets from the last window or continue with the next window transmission based on the CDT NextPacketExpected response. Upon reception of a NextPacketExpected with an invalid value, NGA shall abort the file transfer and close the file with status = FS_WINDOW_ERR.	S3.2.4		SD3.12	TC2.7.1
VAP3.4.12	If the CDT does not respond in time to a Request/Response Ftp_DataWindow packets, or if an UNACKD Ftp_DataWindow packet transmission fails, NGA shall abort the file transfer and close the file with status = FS_TIMEOUT_ERR.	S3.2.4		SD3.12	TC2.7.1
VAP3.4.13	NGA shall indicate the end of transfer by sending an Ftp_DataWindow packets with its Service type = Request/Response and a Length less than the maximum packet size, even if it means transferring an FtpDataWindow packet with its Length = 1.	S3.2.4		SD3.12	TC2.7.1
VAP3.4.14	At the end of the transfer, if the CDT does not send FR_CLOSE_FILE within 10 seconds, NGA shall unitarily close the file with status = FS_TIMEOUT_ERR.	S3.2.4		SD3.12	TC2.7.1
VAP3.5.1	Upon receiving nvi03SCIVersReq, the NGA shall send the nvo03SCIVersion to the CDT with following variables:	S3.2.5		SD3.14 SD3.18	TC2.6.1 TC2.6.2 TC2.6.3 TC2.6.4 TC2.6.5 TC2.6.6
VAP3.5.1.1	Upon receiving sci_index of Boot Loader, NGA shall update: sci_index = 1, sci_type = 5, sci_name = “Boot”, sci_version = [maximum 8 characters], build_date = [YYYY-MM-DD],	S3.2.5		SD3.14 SD3.18	TC2.6.1 TC2.6.2 TC2.6.3 TC2.6.4 TC2.6.5 TC2.6.6
VAP3.5.1.2	Upon receiving sci_index of DCU Application, NGA shall update: sci_index = 2, sci_type = 3, sci_name = “App”, sci_version = [maximum 8 characters], build_date = [YYYY-MM-DD],	S3.2.5		SD3.14 SD3.18	TC2.6.1 TC2.6.2 TC2.6.3 TC2.6.4 TC2.6.5 TC2.6.6
VAP3.5.1.3	Upon receiving sci_index of NGA, NGA shall update: sci_index = 3, sci_type = 3, sci_name = “Neuron”, sci_version = [maximum 8 characters], build_date = [YYYY-MM-DD],	S3.2.5		SD3.14 SD3.18	TC2.6.1 TC2.6.2 TC2.6.3 TC2.6.4 TC2.6.5 TC2.6.6
VAP3.5.1.4	Upon receiving sci_index of CPLD, NGA shall update: sci_index = 2, sci_type = 2, sci_name = “CPLD”, sci_version = [maximum 8 characters], build_date = [Null].	S3.2.5		SD3.14 SD3.18	TC2.6.1 TC2.6.2 TC2.6.3 TC2.6.4 TC2.6.5 TC2.6.6

SRS Reqmt ID	Requirements Description (From SRS)	SRS Ref. Section	SFD / System Ref. ID	SDD Specific Ref. ID	SITP Ref. ID
VAP3.6.1	When NGA receives nvi01TransfState.bit0 = 1, NGA shall stop all transmission of the Object #1 messages.	S3.2.6	SFD1.2	SD3.6	TC2.4.1
VAP3.6.2	NGA shall resume the transmission of all messages of the Object #1 when NGA receives nvi01TransfState. bit0 = 0 or 60s has passed since the last reception of nvi01TransfState.bit0 = 1.	S3.2.6	SFD1.2	SD3.6	TC2.4.1
VAP3.6.3	NGA shall update the nvo01StatusBits (and) with the SPI_Frame.DCUtoNGA.	S3.2.6	SFD1.2	SD3.6	TC2.4.1
VAP3.6.4	By default, NGA shall initially transmit the nvo01StatusBits (and nvo01StatusBits2 if applicable) with the transmission rate of 9s.	S3.2.6	SFD1.2	SD3.6	TC2.4.1
VAP3.6.5	By default, NGA shall not transmit the nvo01Parameter (if applicable).	S3.2.6	SFD1.2	SD3.6	TC2.4.1
VAP3.6.6	When NGA receives nvi01Subscribe with non-zero update_rate:	S3.2.6	SFD1.2	SD3.6	TC2.4.1
VAP3.6.6.1	NGA shall verify the valid range of the update_rate: multiples of 500ms over a range from 0 to 60,000ms.	S3.2.6	SFD1.2	SD3.6	TC2.4.1
VAP3.6.6.2	Upon receiving param_id of zero, NGA shall update the transmission rate of the nvo01StatusBits (and nvo01StatusBits2 if applicable) with the update_rate.	S3.2.6	SFD1.2	SD3.6	TC2.4.1
VAP3.6.6.3	Upon receiving param_id of one, NGA shall update the transmission rate of the nvo01Parameter (if applicable) with the update_rate.	S3.2.6	SFD1.2	SD3.6	TC2.4.1
VAP3.6.7	When NGA receives nvi01Subscribe with zero update_rate:	S3.2.6	SFD1.2	SD3.6	TC2.4.1
VAP3.6.7.1	NGA shall stop the transmission of the nvo01StatusBits (and nvo01StatusBits2 if applicable) for param_id of zero and nvo01Parameter for param_id of one (if applicable).	S3.2.6	SFD1.2	SD3.6	TC2.4.1
VAP3.6.7.2	If the transmission rate of param_id is already zero, then NGA shall send the requested “pram_id” once.	S3.2.6	SFD1.2	SD3.6	TC2.4.1
VAP3.7.1	NGA shall accept the file transfer (respond back with “Initiate OK”) if there is a sufficient memory space, the car is in stop, the version is valid, and doors are closed and locked.	S3.2.7		SD3.5	TC2.7.1
VAP3.7.2	NGA shall respond back to CDT with “Install OK” if the integrity of the newly transferred software image is correct.	S3.2.7		SD3.5	TC2.7.1
VAP3.7.3	NGA shall install transferred software image after responding back to CDT.	S3.2.7		SD3.5	TC2.7.1
VAP3.8.1	Serial synchronous transmission shall occur on SpiSck and SpiMtsr.	S3.2.8		SD3.16	TC2.8.1 TC2.8.2 TC2.8.3 TC2.8.4 TC2.8.5 TC2.8.6 TC2.8.7 TC2.8.8 TC2.8.9 TC2.8.10 TC2.8.11 TC2.8.12
VAP3.8.2	NGA shall assemble SPI_Frame.NGAtoDCU to be transmitted using the latest values at the time the frame is assembled.	S3.2.8		SD3.4 SD3.4.2 SD3.16.3	TC2.8.1 TC2.8.2 TC2.8.3 TC2.8.4 TC2.8.5 TC2.8.6 TC2.8.7 TC2.8.8 TC2.8.9 TC2.8.10 TC2.8.11 TC2.8.12

SRS Reqmt ID	Requirements Description (From SRS)	SRS Ref. Section	SFD / System Ref. ID	SDD Specific Ref. ID	SITP Ref. ID
VAP3.8.3	The transmitted Chksum shall be the sum of the SYNC and Data0 through the Data33 bytes, plus one.	S3.2.8		SD3.4.1 SD3.4.2	TC2.8.1 TC2.8.2 TC2.8.3 TC2.8.4 TC2.8.5 TC2.8.6 TC2.8.7 TC2.8.8 TC2.8.9 TC2.8.10 TC2.8.11 TC2.8.12
VAP3.8.4	NGA shall set the SpiSync frame synchronization signal at the beginning of each frame.	S3.2.8		SD3.4 SD3.16	TC2.8.1 TC2.8.2 TC2.8.3 TC2.8.4 TC2.8.5 TC2.8.6 TC2.8.7 TC2.8.8 TC2.8.9 TC2.8.10 TC2.8.11 TC2.8.12
VAP3.8.5	NGA shall periodically send (and thus receive) a frame every 100ms ± 30ms.	S3.2.8		SD3.4.1 SD3.4.2	TC2.8.1 TC2.8.2 TC2.8.3 TC2.8.4 TC2.8.5 TC2.8.6 TC2.8.7 TC2.8.8 TC2.8.9 TC2.8.10 TC2.8.11 TC2.8.12
VAP3.8.6	NGA shall decompose the incoming synchronous stream of bytes SPI_Frame.DCUtoNGA.	S3.2.8		SD3.4.1 SD3.4.2 SD3.16.1	TC2.8.1 TC2.8.2 TC2.8.3 TC2.8.4 TC2.8.5 TC2.8.6 TC2.8.7 TC2.8.8 TC2.8.9 TC2.8.10 TC2.8.11 TC2.8.12

SRS Reqmt ID	Requirements Description (From SRS)	SRS Ref. Section	SFD / System Ref. ID	SDD Specific Ref. ID	SITP Ref. ID
VAP3.8.7	Upon reception of a frame with a correct SYNC byte and a correct CHK7:0, NGA shall decompose the data part of the frame. A correct CHK7: 0 should be the summation of the SYNC and Data bytes, plus one.	S3.2.8		SD3.4.1 SD3.4.2	TC2.8.1 TC2.8.2 TC2.8.3 TC2.8.4 TC2.8.5 TC2.8.6 TC2.8.7 TC2.8.8 TC2.8.9 TC2.8.10 TC2.8.11 TC2.8.12
VAP3.8.8	Given a frame that does not comply with the SPI_Frame.DCUtoNGA, the data frame shall be disregarded.	S3.2.8		SD3.4.1 SD3.4.2	TC2.8.1 TC2.8.2 TC2.8.3 TC2.8.4 TC2.8.5 TC2.8.6 TC2.8.7 TC2.8.8 TC2.8.9 TC2.8.10 TC2.8.11 TC2.8.12
VAP3.8.9	Given a frame that does not comply with the SPI_Frame.DCUtoNGA, NGA shall set the SPIErr of the SPI_Frame.NGAtoDCU to 1, otherwise to 0. The updated status will thus be sent in the next frame.	S3.2.8		SD3.4.2	TC2.8.1 TC2.8.2 TC2.8.3 TC2.8.4 TC2.8.5 TC2.8.6 TC2.8.7 TC2.8.8 TC2.8.9 TC2.8.10 TC2.8.11 TC2.8.12
VAP3.8.10	If the SPIErr has been 1 for 2.0 seconds consecutively, the SPI Interface status is “bad”, otherwise it is “ok”. Upon NGA startup, the initial status of SPI shall be set to “bad”.	S3.2.8		SD3.1 SD3.4.2	TC2.8.1 TC2.8.2 TC2.8.3 TC2.8.4 TC2.8.5 TC2.8.6 TC2.8.7 TC2.8.8 TC2.8.9 TC2.8.10 TC2.8.11 TC2.8.12

SRS Reqmt ID	Requirements Description (From SRS)	SRS Ref. Section	SFD / System Ref. ID	SDD Specific Ref. ID	SITP Ref. ID
VAP3.8.11	Upon reception of valid SPI frames with the valid ExtraDataID in consecutive ascending order of ExtraDataID, NGA shall decompose the ExtraData.	S3.2.8		SD3.4.2 SD3.16.1	TC2.8.1 TC2.8.2 TC2.8.3 TC2.8.4 TC2.8.5 TC2.8.6 TC2.8.7 TC2.8.8 TC2.8.9 TC2.8.10 TC2.8.11 TC2.8.12
VAP3.9.1	Upon startup, the No Lan Communication (NLC) shall be 1 by default.	S3.2.9		SD3.1	TC2.5.1
VAP3.9.2	The NLC shall be 1 when NGA detects the network comunication is down for 60s.	S3.2.9		SD3.2 SD3.7	TC2.5.1
VAP3.9.3	The NLC shall be 0 when NGA detects the network communication is up again.	S3.2.9		SD3.2 SD3.7	TC2.5.1
VAP3.9.4	The LonWorksErr shall be 0, when there is no Firmware error and 1 when there is.	S3.2.9		SD3.2 SD3.7	TC2.5.1
VAP3.10.1	At startup, NGA shall identify its location by using its Discrete input interface.	S3.2.10		SD3.1	TC2.1.1 TC2.1.2 TC2.1.3 TC2.1.4
VAP3.10.2	NGA shall identify its location at startup only, ignoring changes that could occur later.	S3.2.10		SD3.1	TC2.1.1 TC2.1.2 TC2.1.3 TC2.1.4
VAP3.10.3	For valid configurations, NGA shall set the DCU node HI selectors of the Network Variable Configuration as following: DCU1:4 to 0x11;0x14 according to the location read from IDENT1 and IDENT2.	S3.2.10		SD3.1	TC2.1.1 TC2.1.2 TC2.1.3 TC2.1.4
VAP3.10.4	Upon invalid configuration, NGA shall not transmit any network objects.	S3.2.10		SD3.1	TC2.1.1 TC2.1.2 TC2.1.3 TC2.1.4
VAP3.11.1	NGA performs self-tests at startup:	S3.2.11		SD3.1	TC2.2.1 TC2.3
VAP3.11.1.1	NGA shall compute the CRC of the application code, and compare to the stored CRC in NGA_Crc15:0 of the CRC Database Fields.	S3.2.11		SD3.3	TC2.2.1 TC2.3
VAP3.11.1.2	NGA shall verify that the Echelon FT3150 chip register functions correctly.	S3.2.11		SD3.3	TC2.2.1 TC2.3
VAP3.11.2	Upon failure of a self-test, NGA shall place the Neuron in an OFF-LINE state, turning off its discreet outputs, preventing all communications with the SPI, and preventing all high-level functions with the car network.	S3.2.11		SD3.3	TC2.2.1 TC2.3
VAP3.11.3	If all tests pass, NGA shall report its CRC in NGA_Crc (SPI_Frame.NGAtoDCU), and its software version in NGA_VerMaj and NGA_VerMin (SPI_Frame.NGAtoDCU).	S3.2.11		SD3.3 SD3.4.2	TC2.2.1 TC2.3
VAP3.12.1	NGA shall access the car network through Rx, Tx, and TxEnable of the Network Hardware Interface in single-ended mode at 78.1kbps nominal rate, where the Cdet# of the Network Hardware Interface functions as a collision detection input (however, Cdet# is ignored as the cd_t of the TransceiverParameters Data Fields is set to 0).	S3.2.12		SD3.1	TC2.1.1 TC2.1.2 TC2.1.3 TC2.1.4
VAP3.12.2	The Sleep output of the Network Hardware Interface is not used by the DCU and can be driven by the NGA in any way it sees.	S3.2.12		SD3.1	TC2.1.1 TC2.1.2 TC2.1.3 TC2.1.4

SRS Reqmt ID	Requirements Description (From SRS)	SRS Ref. Section	SFD / System Ref. ID	SDD Specific Ref. ID	SITP Ref. ID
VAP3.12.3	NGA shall set the TransceiverParameters Data Fields to the exact values specified in Table 3-6.	S3.2.12		SD3.1	TC2.1.1 TC2.1.2 TC2.1.3 TC2.1.4
VAP3.13.1	NGA shall set the config_source of the SCPTnwrkCnfg to the CFG_LOCAL initially.	S3.2.13		SD3.1	TC2.9.1

Note 2: Requirement traceability link will be added during the “SCT-600 Software Coding and Testing” phase.